MARINE COMMUNITY ECOLOGY I

Housekeeping

- Self-assessments
- Project 2
- Resubmissions of Project 1
- Syllabus revision: labs & lectures


DRIFT

SELECTION

DISPERSAL

SPECIATION


DISPERSAL

2 QUESTIONS

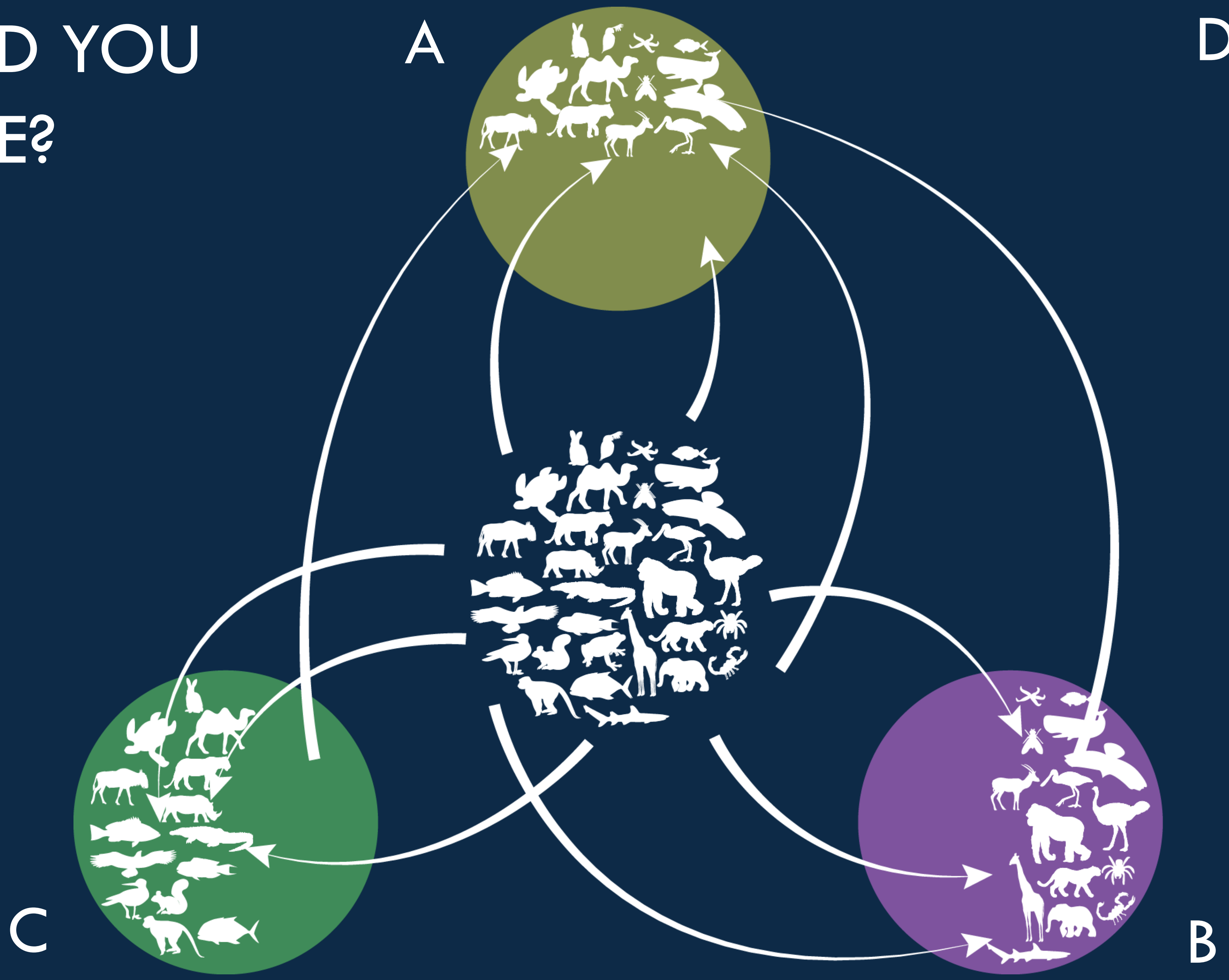
How did you get here?

Where did you come from?


HOW DID YOU GET HERE?

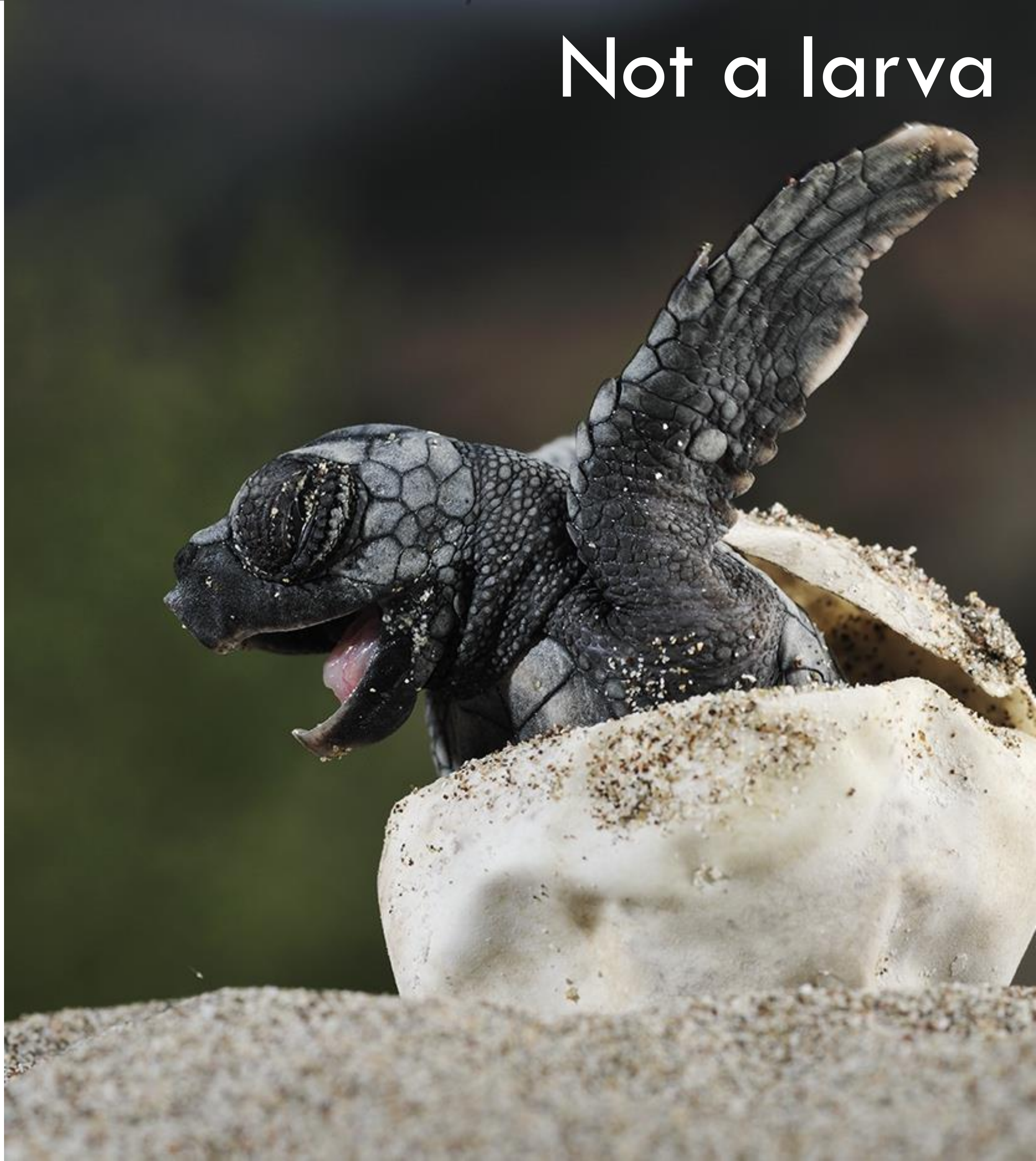
HOW DID YOU GET HERE?

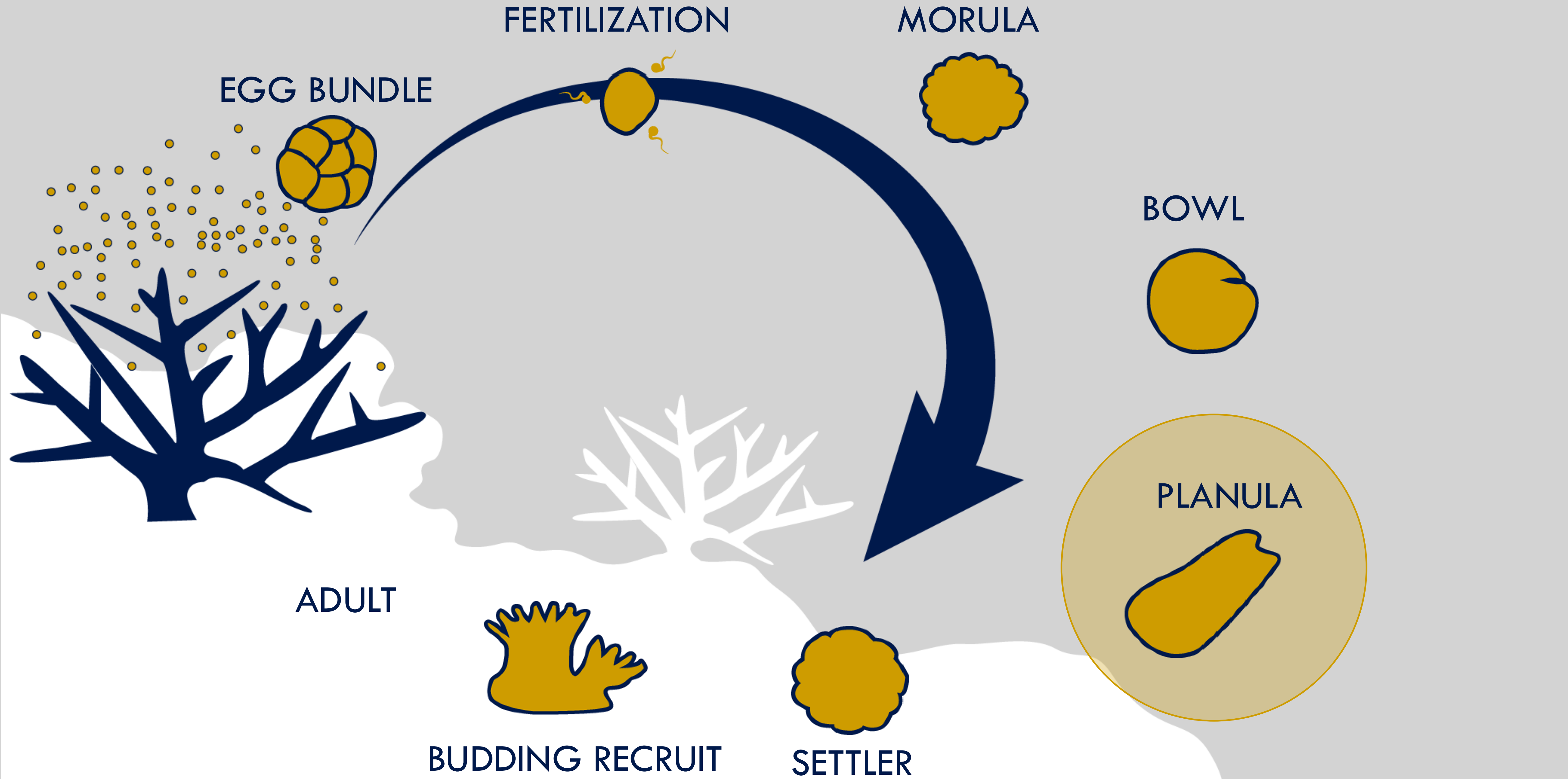
DISPERSAL



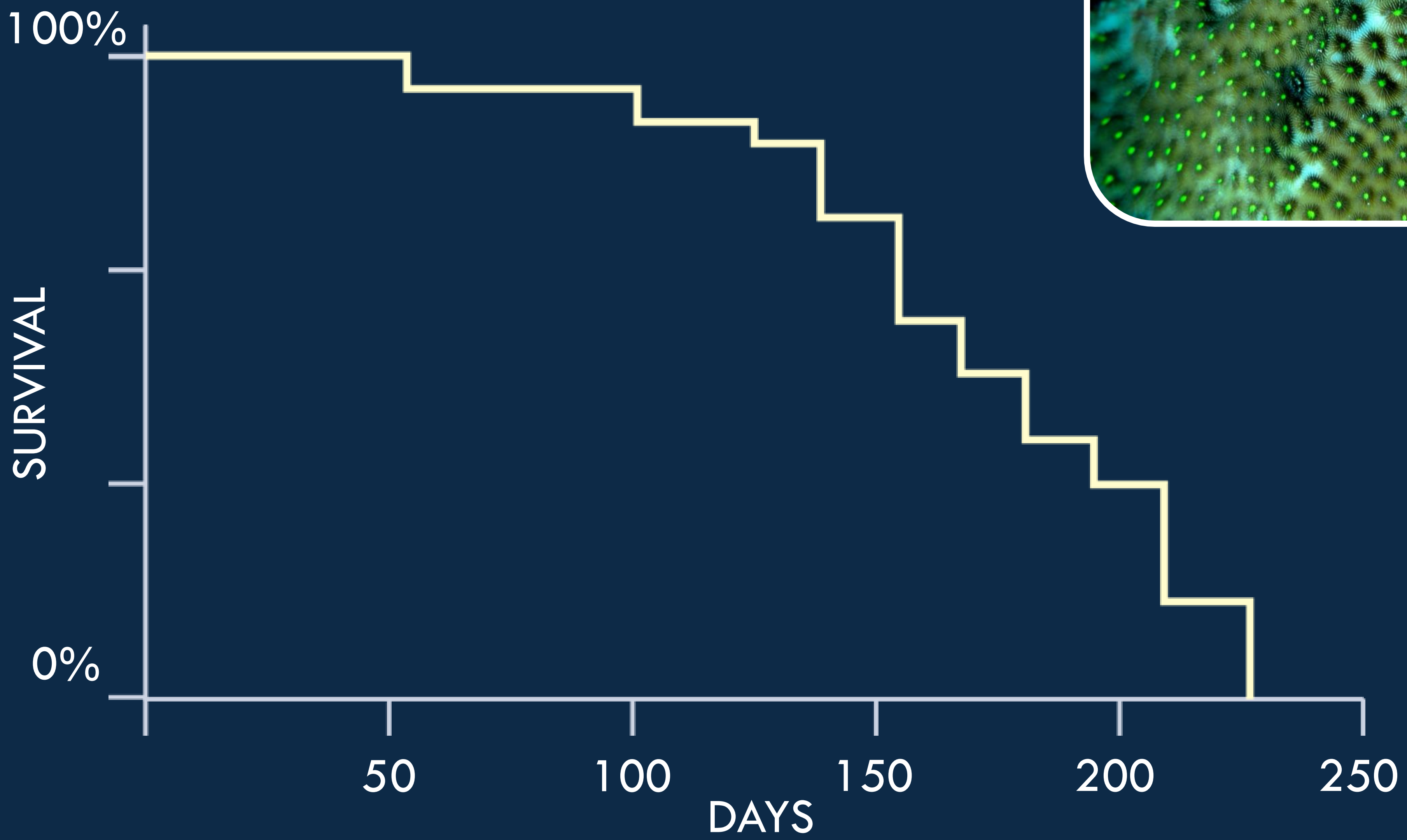

WHERE DID YOU COME FROM?

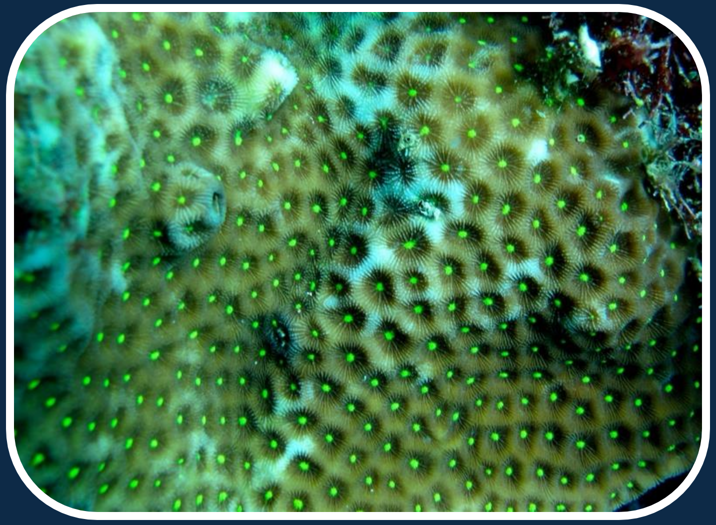


Larva

Not a larva


FERTILIZATION
MORULA
EGG BUNDLE
BOWL
PLANULA
ADULT
BUDDING RECRUIT
SETTLER

Pelagic larval duration






HOW FAR CAN THEY GO?

200 days

Average ocean surface current: 25cm/second

86,400 s/day

2,160,000 cm/day → 21.6km/day

4,320 km dispersal potential


Distance
4,351.55 km
Kyrgyzstan
China
Shanghai
上海
Busan
부산
Osaka
大阪
Tokyo
東京
New Delhi
नई दिल्ली
Kolkata
কলকাতা
Mumbai
मुंबई
Taipei
台北
Taiwan
Laos
Hong Kong
香港
Philippine Sea
North
Pacific
Ocean
South
China Sea
Luzon
Bangkok
Vietnam
Philippines
Bengaluru
Gulf of
Thailand
Palawan
Mindanao
Basilan Island
Laccadive Sea
Kuala Lumpur
Celebes Sea
Singapore
Jakarta
Surabaya
Indonesia
Bismarck Sea
Papua New
Guinea
Arafura Sea
Solomon Sea
Timor Sea
Suva
Indian
Ocean
Coral Sea
NORTHERN
TERRITORY
QUEENSLAND
Brisbane
WESTERN
AUSTRALIA
Australia
Perth
Sydney
Auckland
Adelaide
Great
Australian
Bight
ACT
VICTORIA
New
3D


ALBERTA
MANITOBA
Labrador Sea
London
Paris
NEWFOUNDLAND AND LABRADOR
France
Seattle
ONTARIO
QUEBEC
Distance
4,314.59 km
MONTANA
Madrid
Lisbon
MINNESOTA
Montreal
OREGON
Toronto
PE
WYOMING
MAINE
Chicago
Ottawa
NEVADA
NEBRASKA
ILLINOIS
RI
COLORADO
PENN
Los Angeles
MISSOURI
WEST VIRGINIA
New York
Start ne
Las Vegas
Morocco
San Diego
OKLAHOMA
North Atlantic Ocean
NEW MEXICO
TENNESSEE
MISSISSIPPI
SOUTH CAROLINA
TEXAS
Gulf of California
Western Sahara
LOUISIANA
Houston
FLORIDA
Gulf of Mexico
Guadalajara
Mexico City
Dakar
Cuba
Santo Domingo
Liberia
Guatemala
Caribbean Sea
Nicaragua
Caracas
Costa Rica
French Guiana
Venezuela
Bogotá
Guyana
Medellín
STATE OF AMAPÁ
STATE OF RORAIMA
Quito
STATE OF PARAÍBA
Ecuador
STATE OF PARÁ
STATE OF SERGIPE
STATE OF AMAZONAS
Brazil
3D
STATE OF ACRE
STATE OF RONDÔNIA
STATE OF ESPÍRITO SANTO
Belo Horizonte
Hanga Roa
Lima
Asunción
São Paulo
100%
Google Data SIO, NOAA, U.S. Navy, NGA, GEBCO Landsat / Copernicus IBCAO INEGI U.S. Geological Survey PGC/NASA
STATE OF RIO GRANDE
1,000 km
Camera: 12,618 km 20°09'36"N 79°16


HOW FAR DO THEY GO?

Organismal dispersal traits

- fertilization
- gamete release
- larval type

External vs. internal fertilization

Broadcast vs. demersal vs. brooding

Planktotrophic vs. lecithotrophic vs. aplanktonic

Planktotrophic larvae

- deplete yolk sac quickly (within hours to days)
- can actively feed on planktonic organisms

Lecithotrophic larvae

"The lecithotrophic pelagic larvae, mainly developing from fairly large yolky eggs, are of a clumsy shape, rather unfit for locomotion"

- require higher maternal investment
- dispersal times are generally shorter

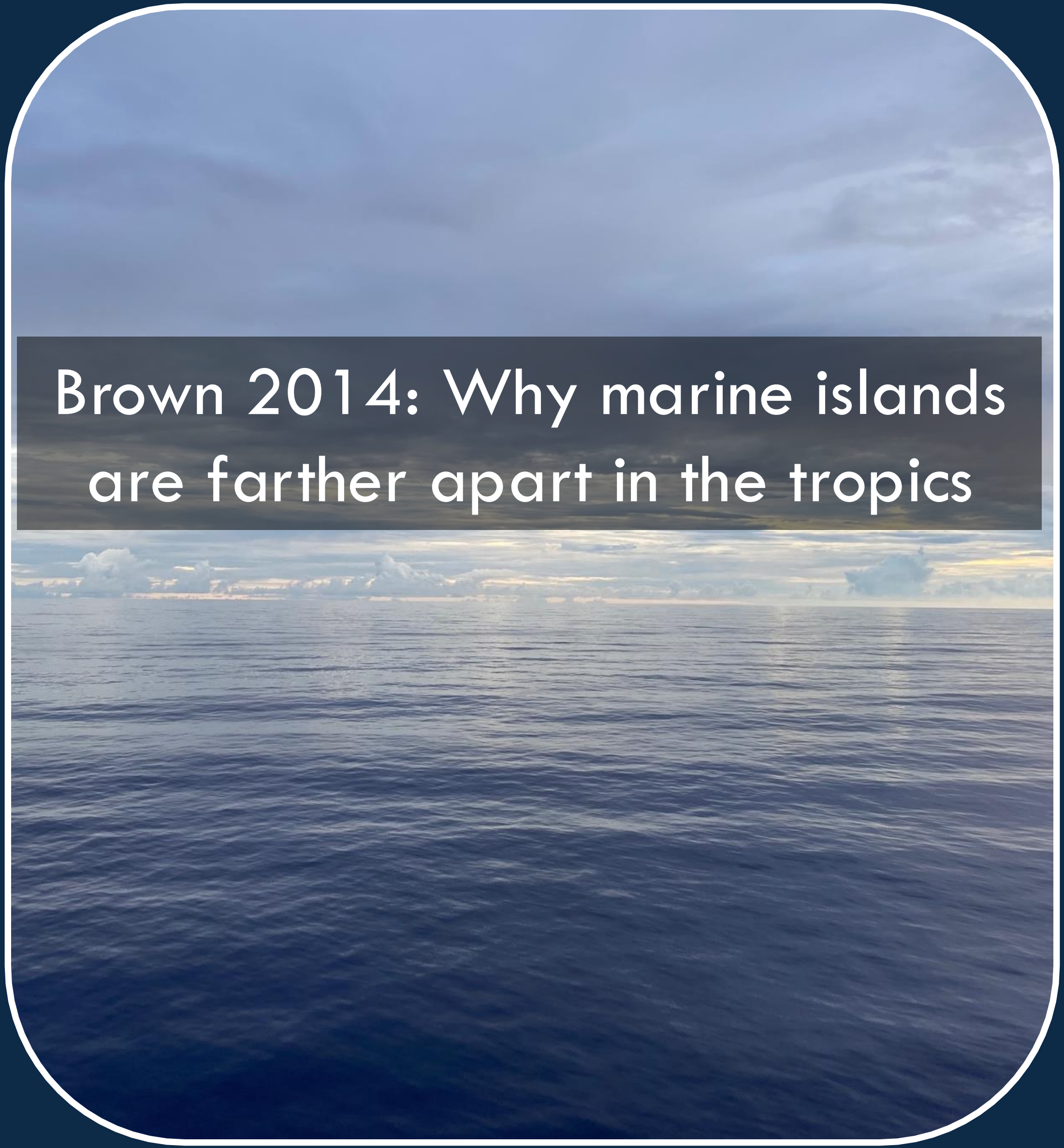
Thorson 1950, Biol Rev

Aplanktonic larvae

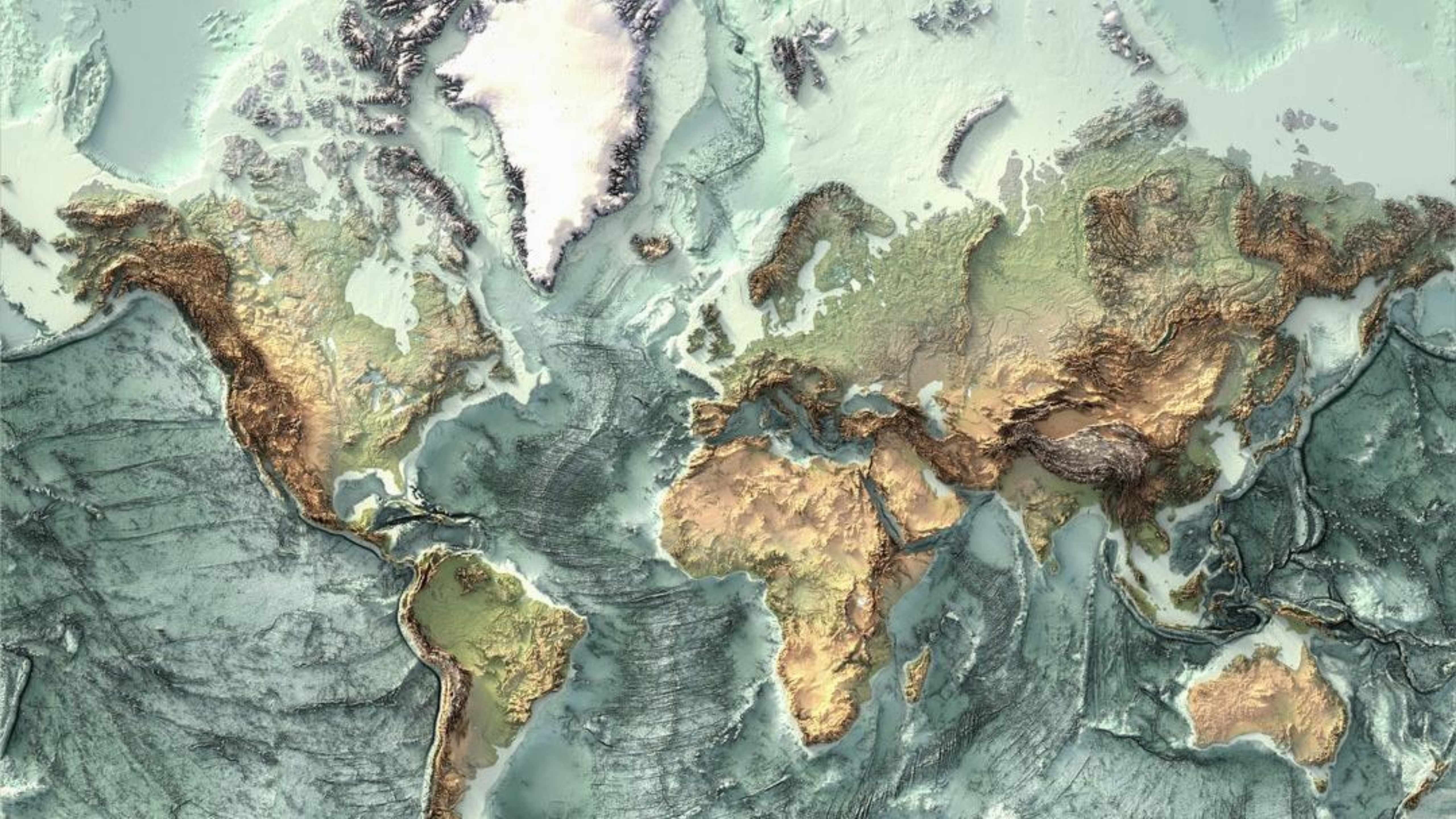
- not really larvae
- hatch fully developed from egg capsule or parent
- viviparous, ovoviviparous, oviparous
- less opportunity to disperse

Organismal dispersal potential

Janzen 1967: Why mountain passes are higher in the tropics



Brown 2014: Why marine islands are farther apart in the tropics




SOOO COLD

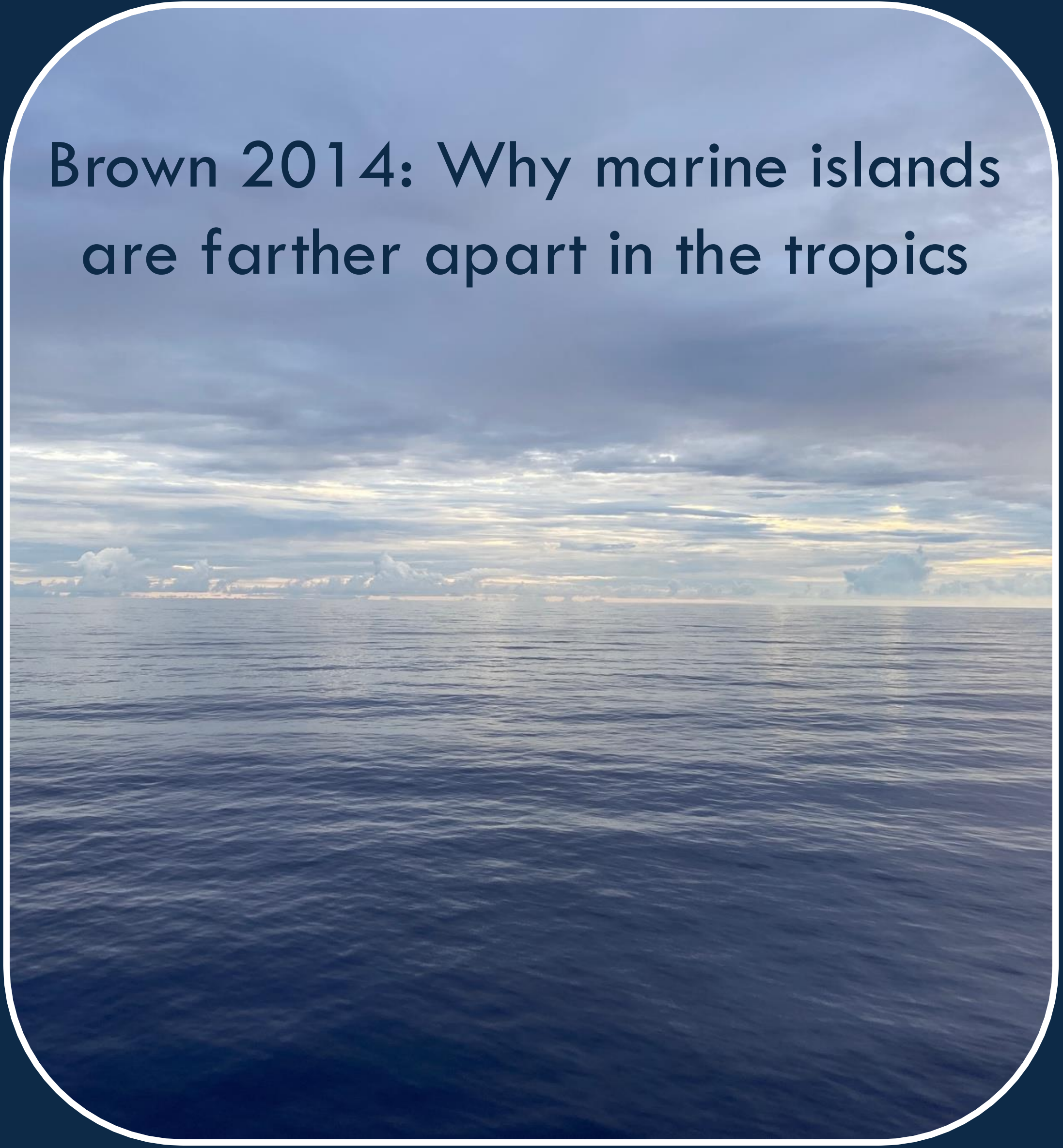

I DON'T WANT TO SAY IT'S HOT
IN MY ROOM
BUT TWO HOBBITS JUST CAME
AROUND AND THREW A RING IN IT

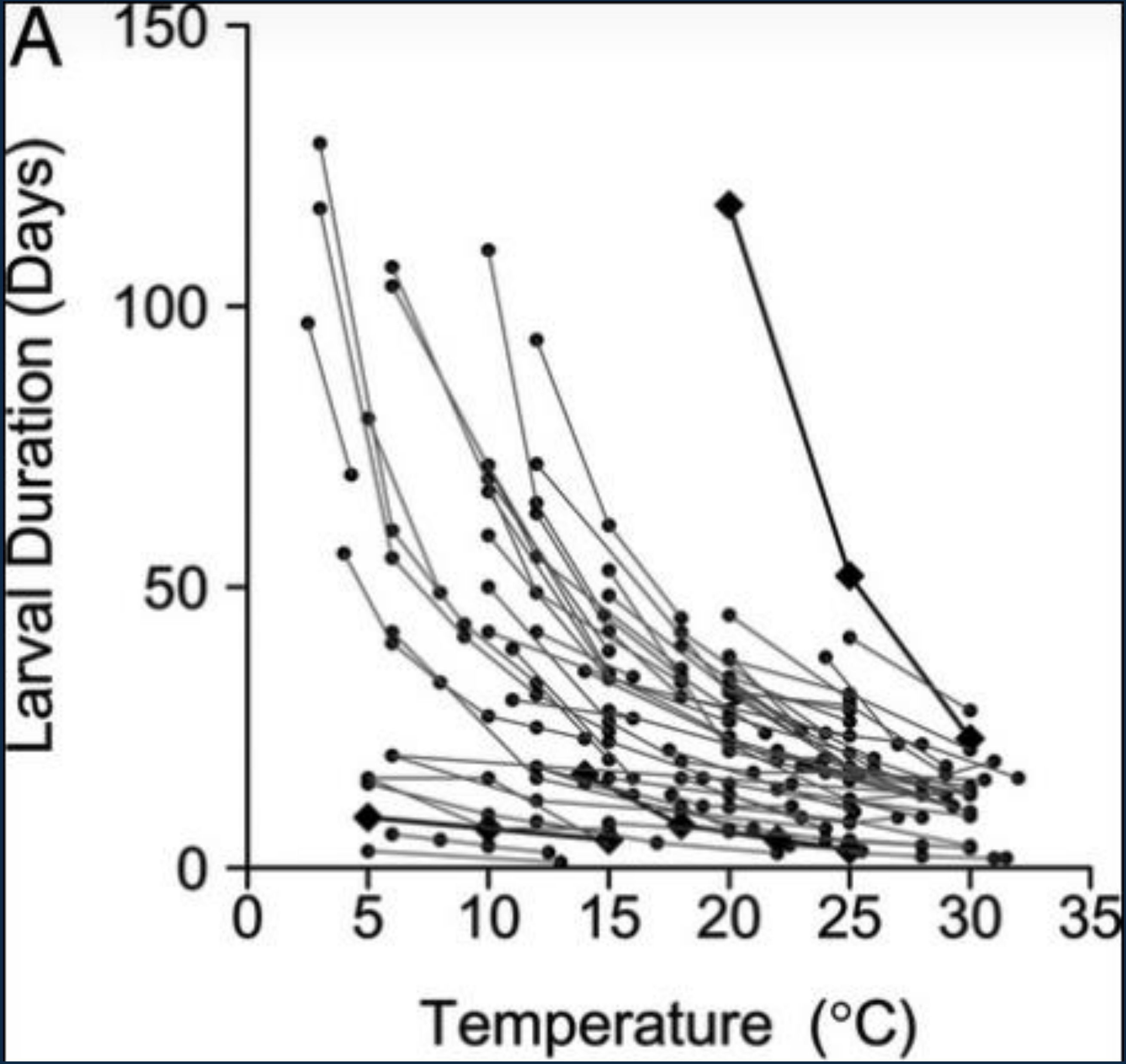
Discussion

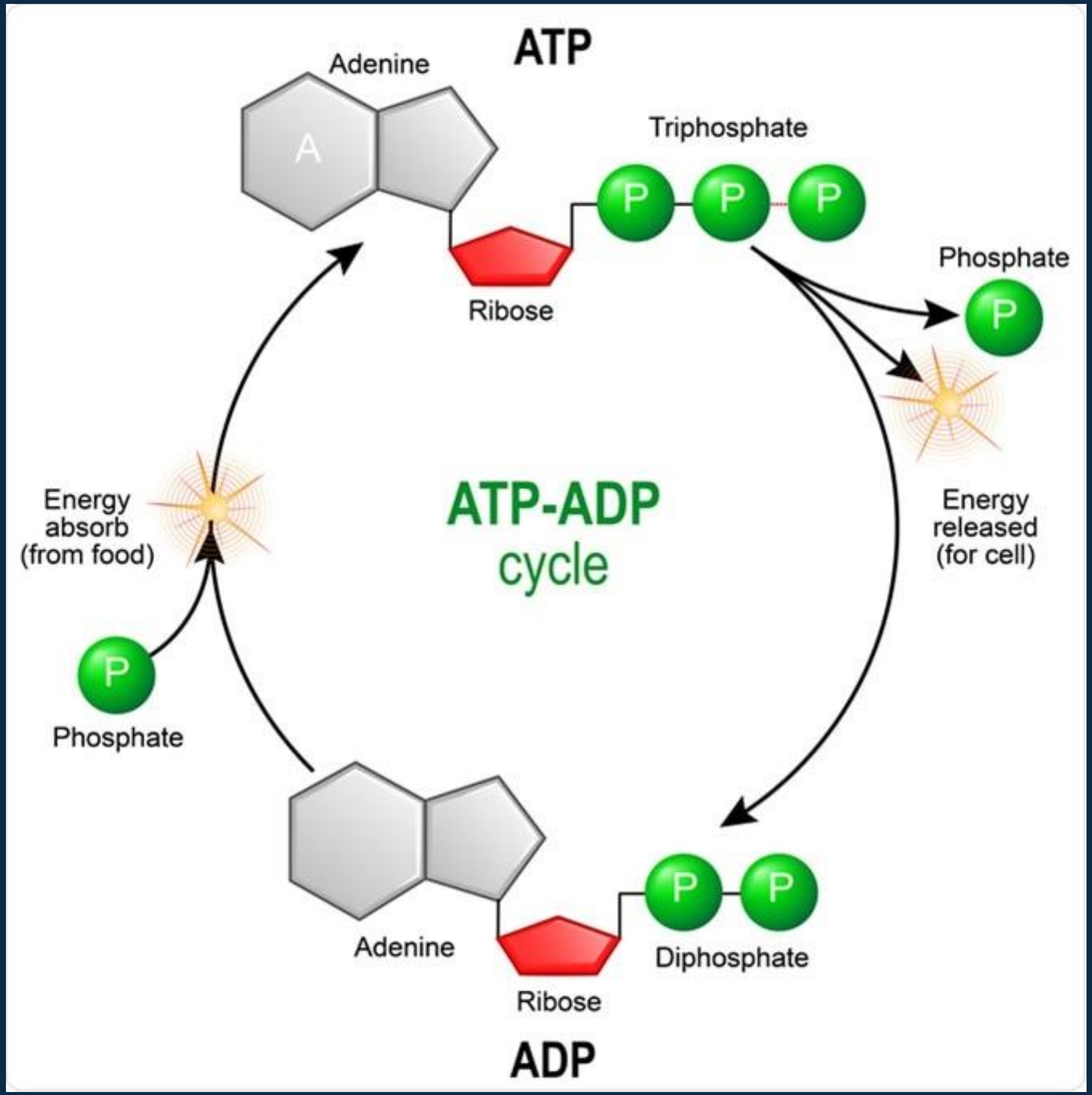
Take 5 minutes to discuss

1) What does Brown mean and what is the difference between terrestrial and marine systems?

2) If mountains are higher and islands farther apart in the tropics, how would this shape global community structure?

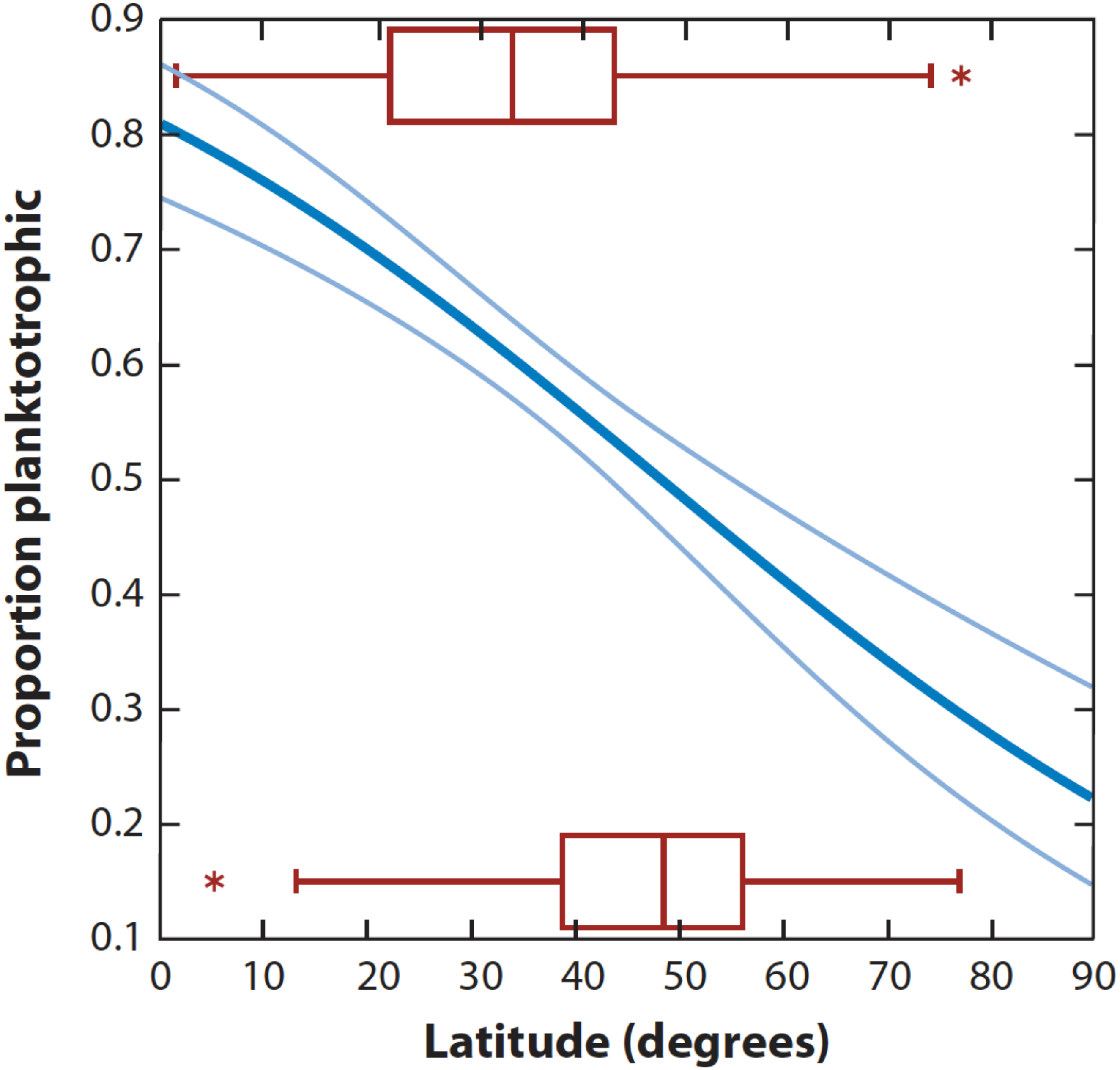






O'Connor et al. 2007, PNAS

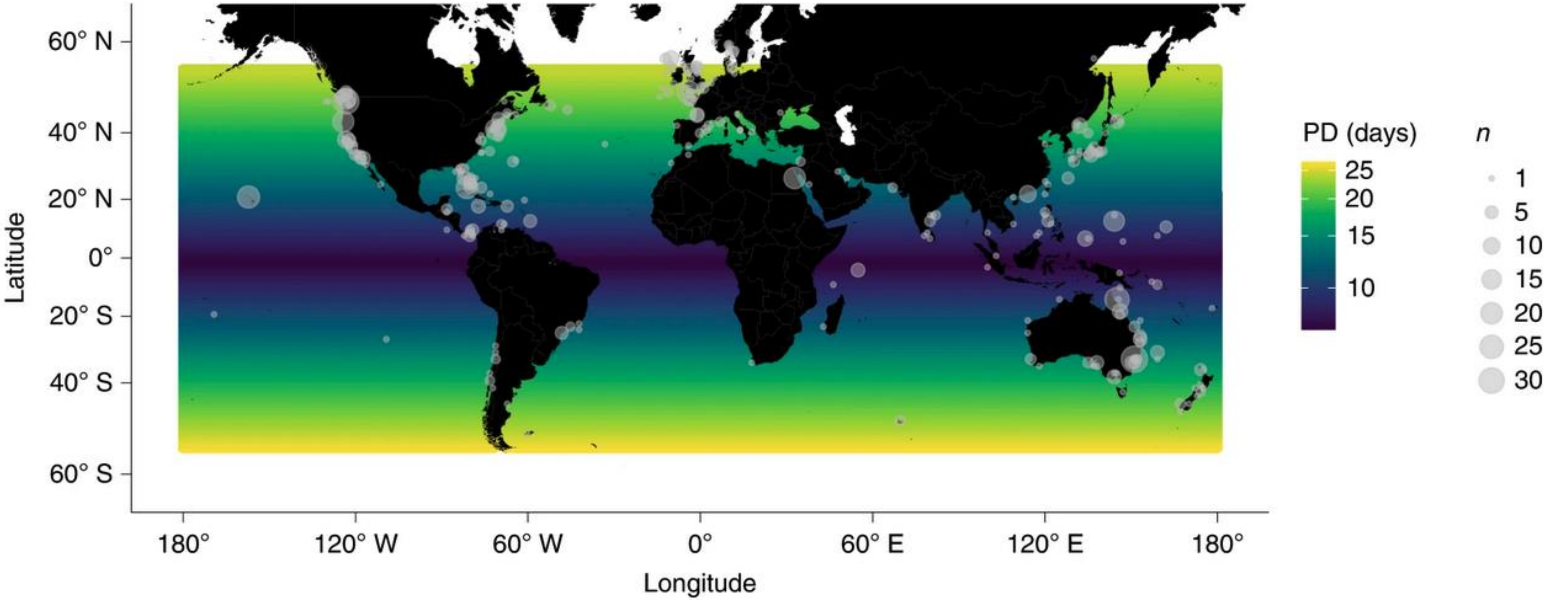
Thorson's rule



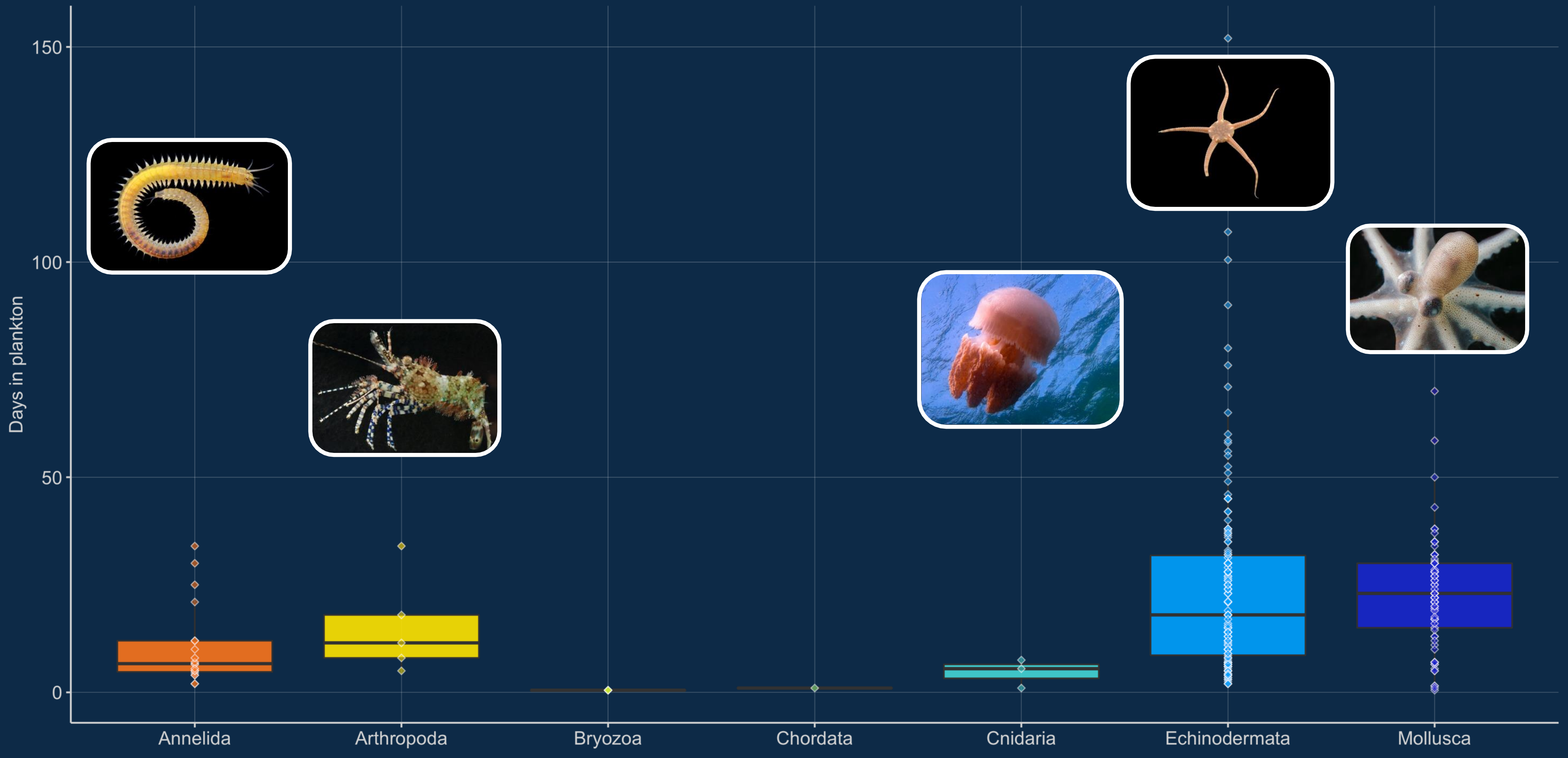
Marshall et al. 2012, AREES

Longer development because of the cold,
but fewer planktotrophic larvae

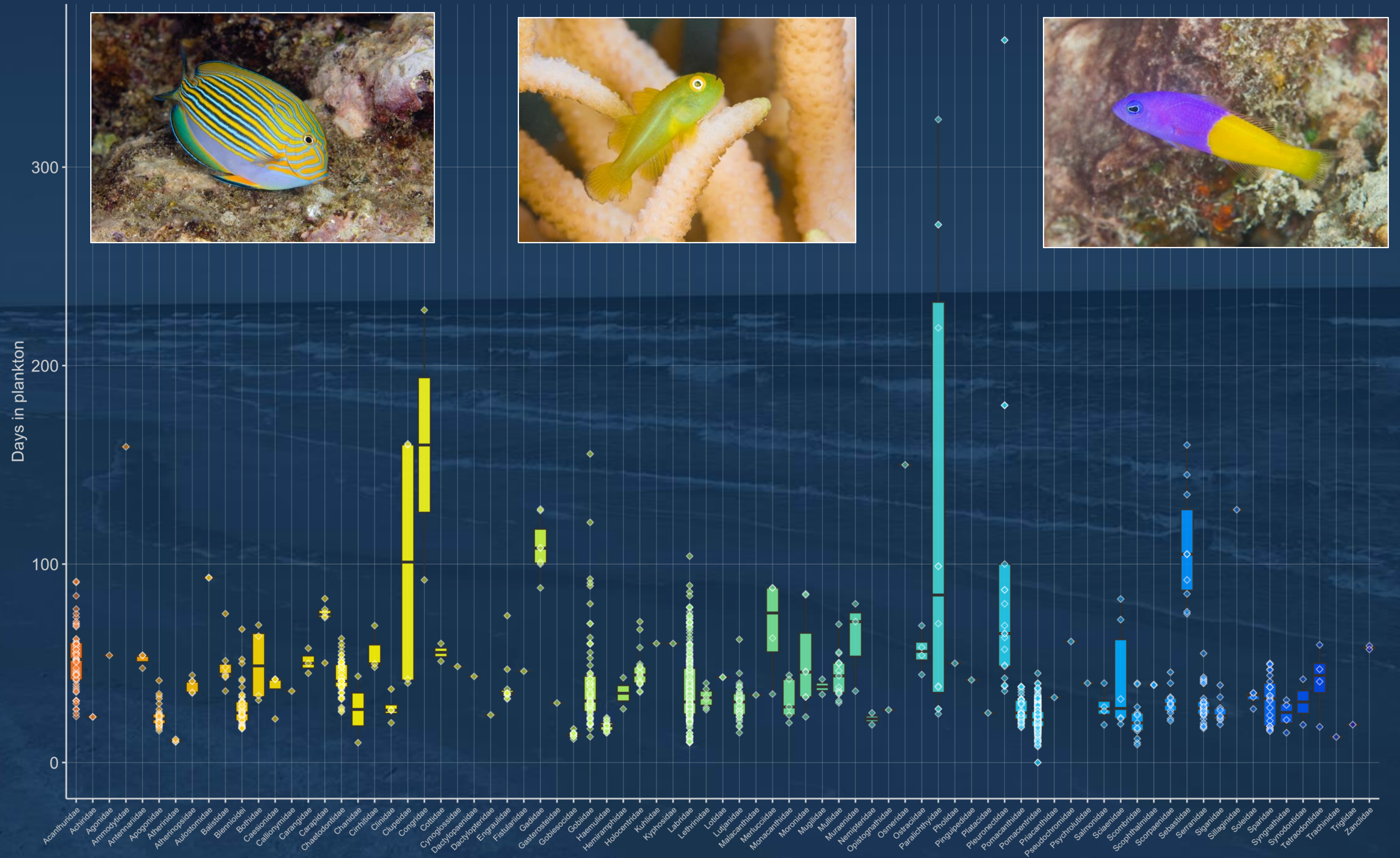
Conflicting impacts of environmental factors vs organismal traits

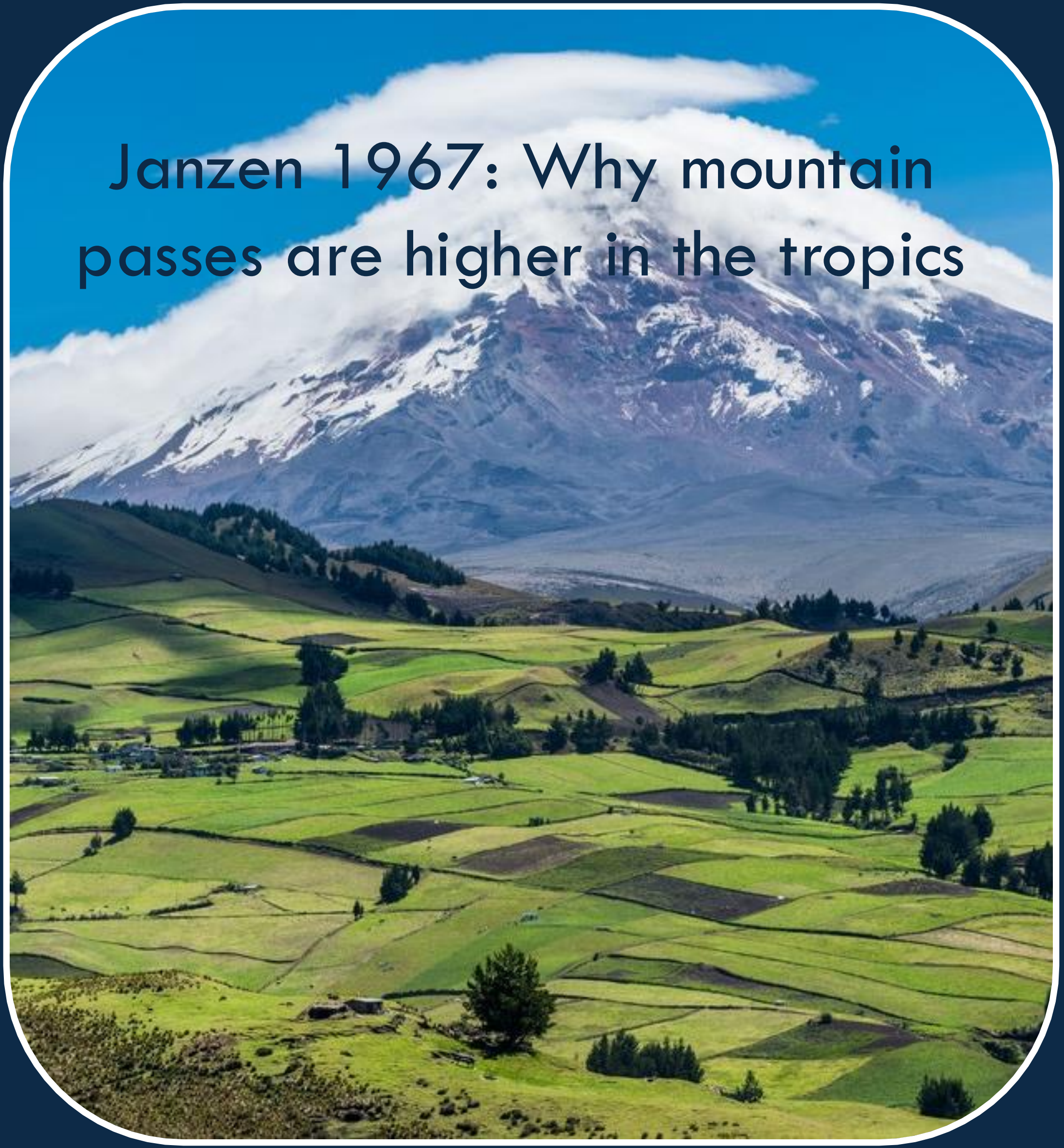


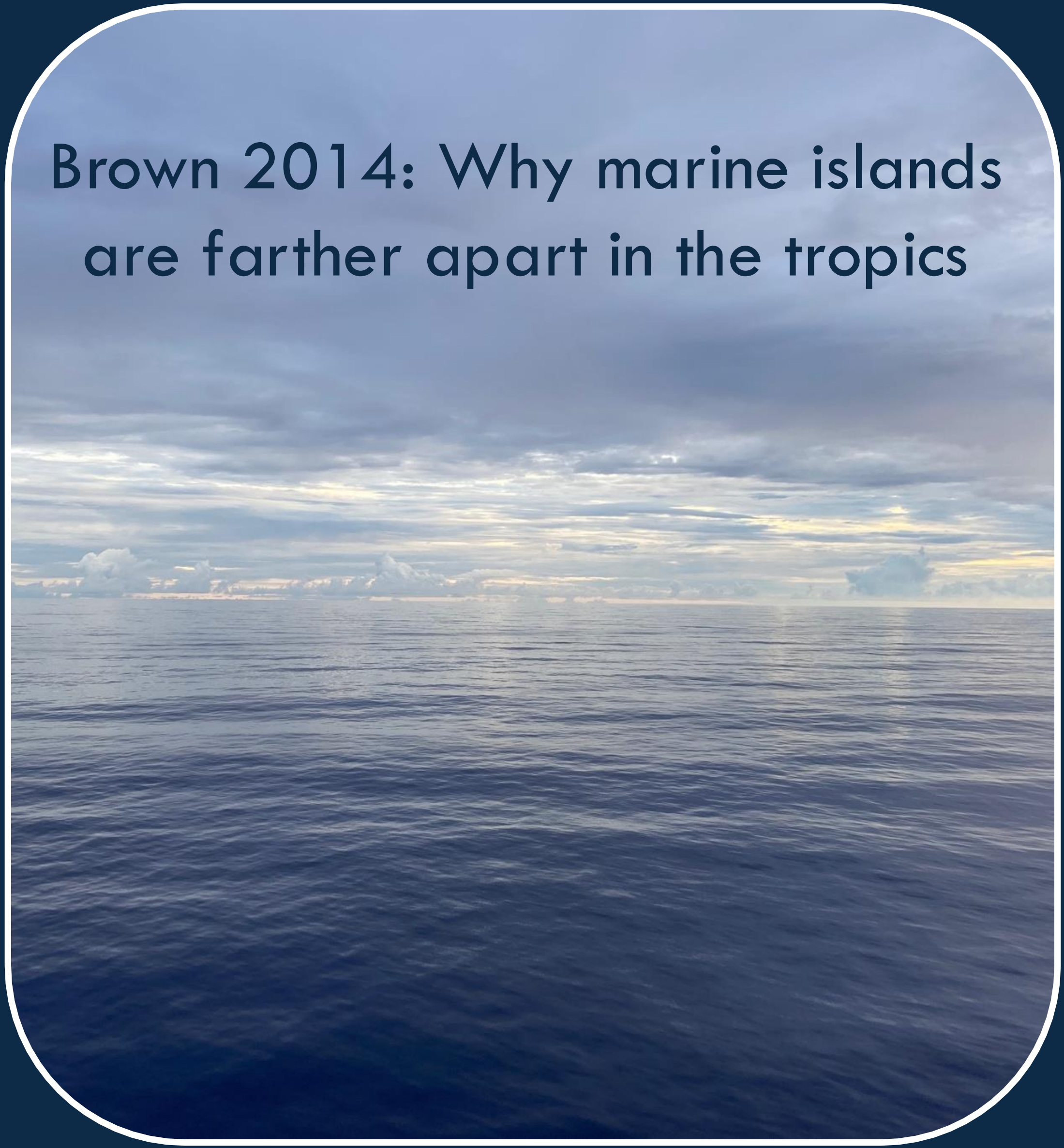
Alvarez-Noriega et al. 2020, Nature Eco Evo

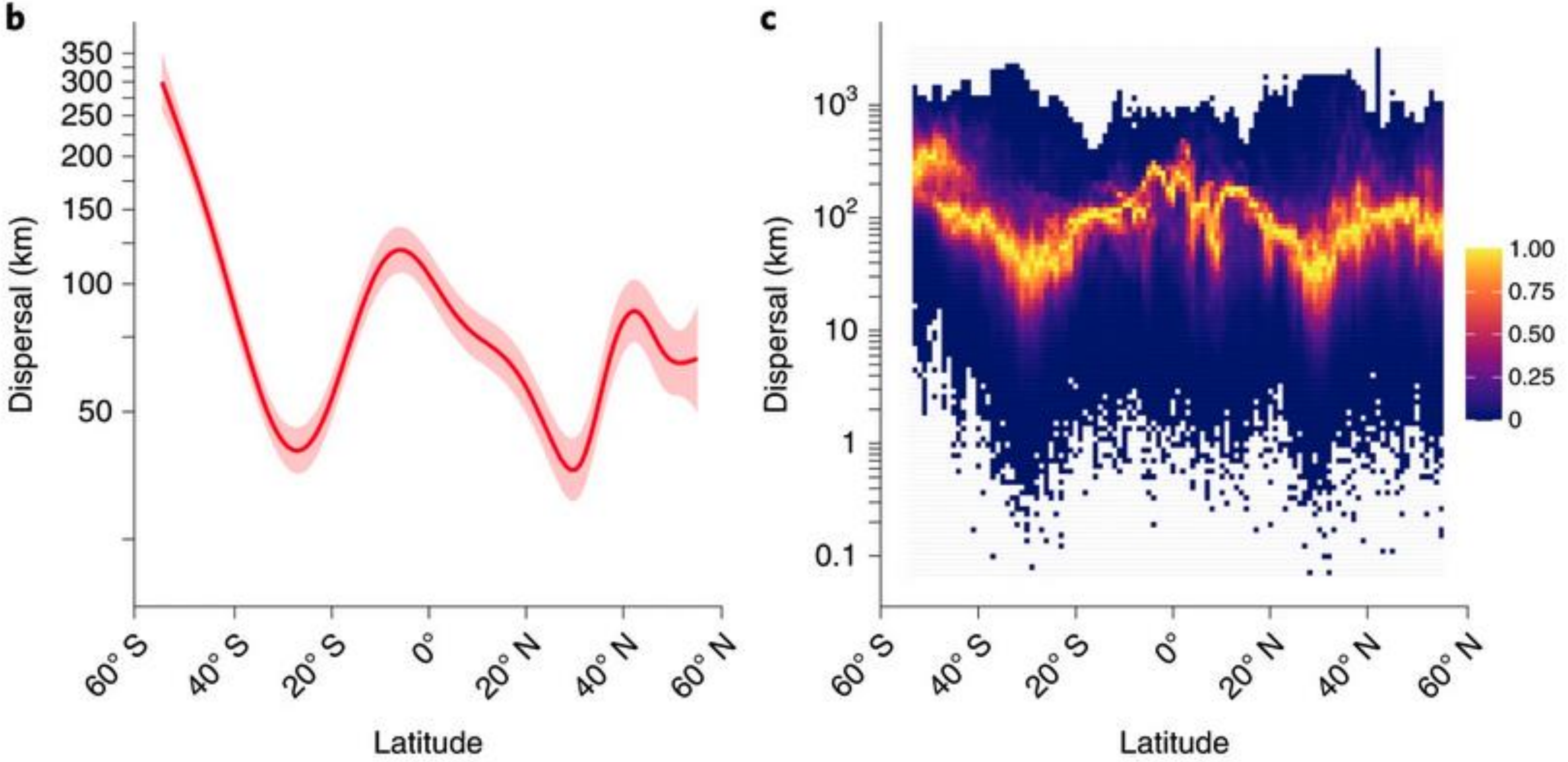


150
100
50
0
Days in plankton
Annelida
Arthropoda
Bryozoa
Chordata
Cnidaria
Echinodermata
Mollusca


Days in plankton
300
200
100
0
Acanthuridae
Achiridae
Agonidae
Ammodytidae
Antennariidae
Apogonidae
Atherinidae
Atherinopsidae
Aulostomidae
Balistidae
Blennioidei
Bothidae
Caesionidae
Callionymidae
Carangidae
Carapidae
Chaetodontidae
Chanidae
Cirrhitidae
Clinidae
Clupeidae
Congridae
Cottidae
Cynoglossidae
Dactylopaenidae
Dactylopteridae
Engraulidae
Fistulariidae
Gadidae
Gasterosteidae
Gobiesocidae
Gobiidae
Haemulidae
Hemiramphidae
Holocentridae
Kuhliidae
Kyphosidae
Labridae
Lethrinidae
Lotidae
Lutjanidae
Malacanthidae
Merlucciidae
Monacanthidae
Moronidae
Mugilidae
Mullidae
Muraenidae
Nemipteridae
Opistognathidae
Osmeridae
Ostraciidae
Paralichthyidae
Pholidae
Pinguipedidae
Platacidae
Pleuronectidae
Pomacanthidae
Pomacentridae
Priacanthidae
Pseudochromidae
Psychrolutidae
Salmonidae
Sciaenidae
Scombridae
Scophthalmidae
Scorpaenidae
Sebastidae
Serranidae
Siganidae
Sillaginidae
Soleidae
Sparidae
Syngnathidae
Synodontidae
Tetraodontidae
Trachinidae
Triglidae
Zanclidae


Janzen 1967: Why mountain passes are higher in the tropics


Brown 2014: Why marine islands are farther apart in the tropics



Alvarez-Noriega et al. 2020, Nature Eco Evo

Longer planktonic duration

≠

Longer dispersal

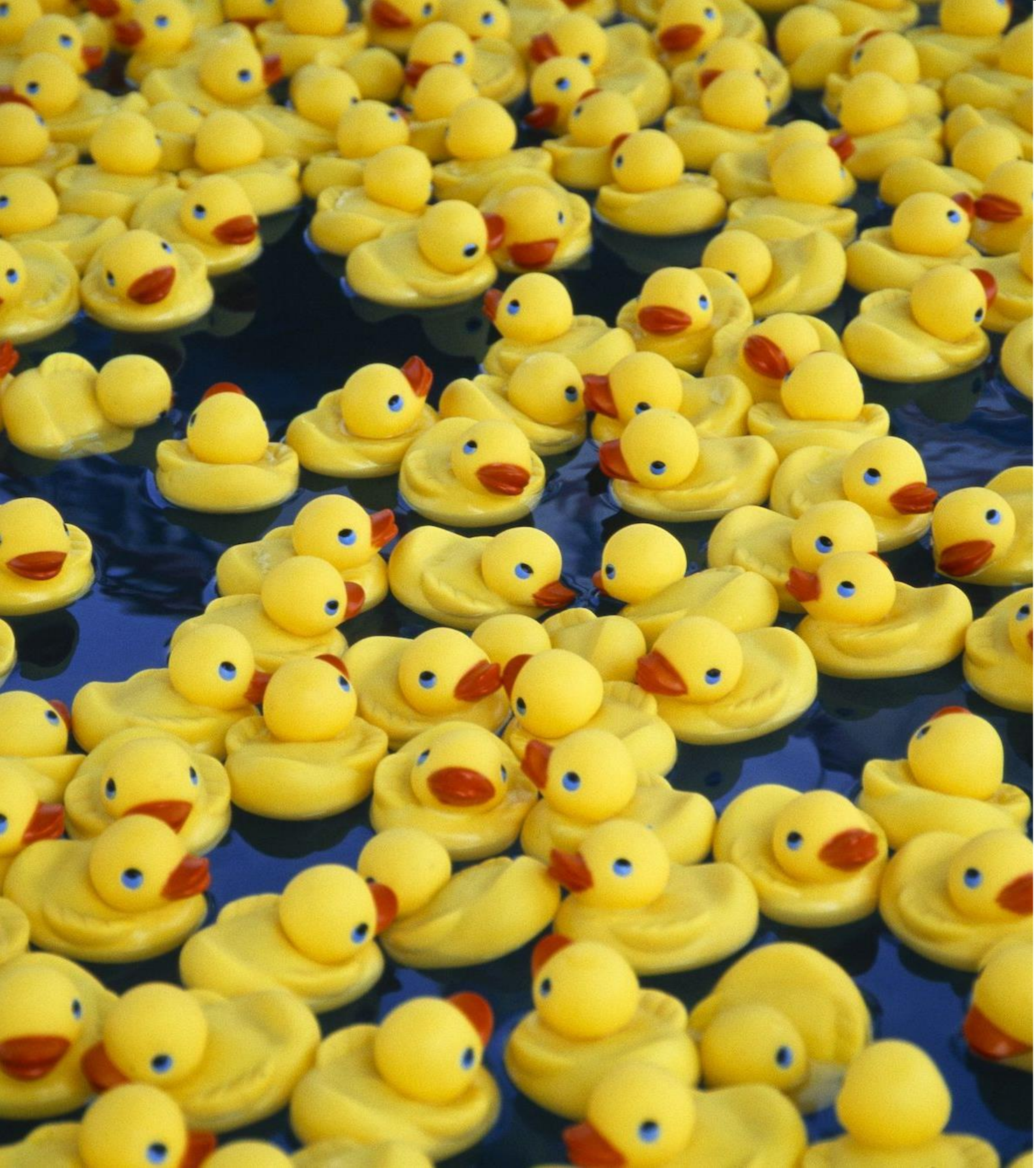
Dispersal potential

Actual dispersal

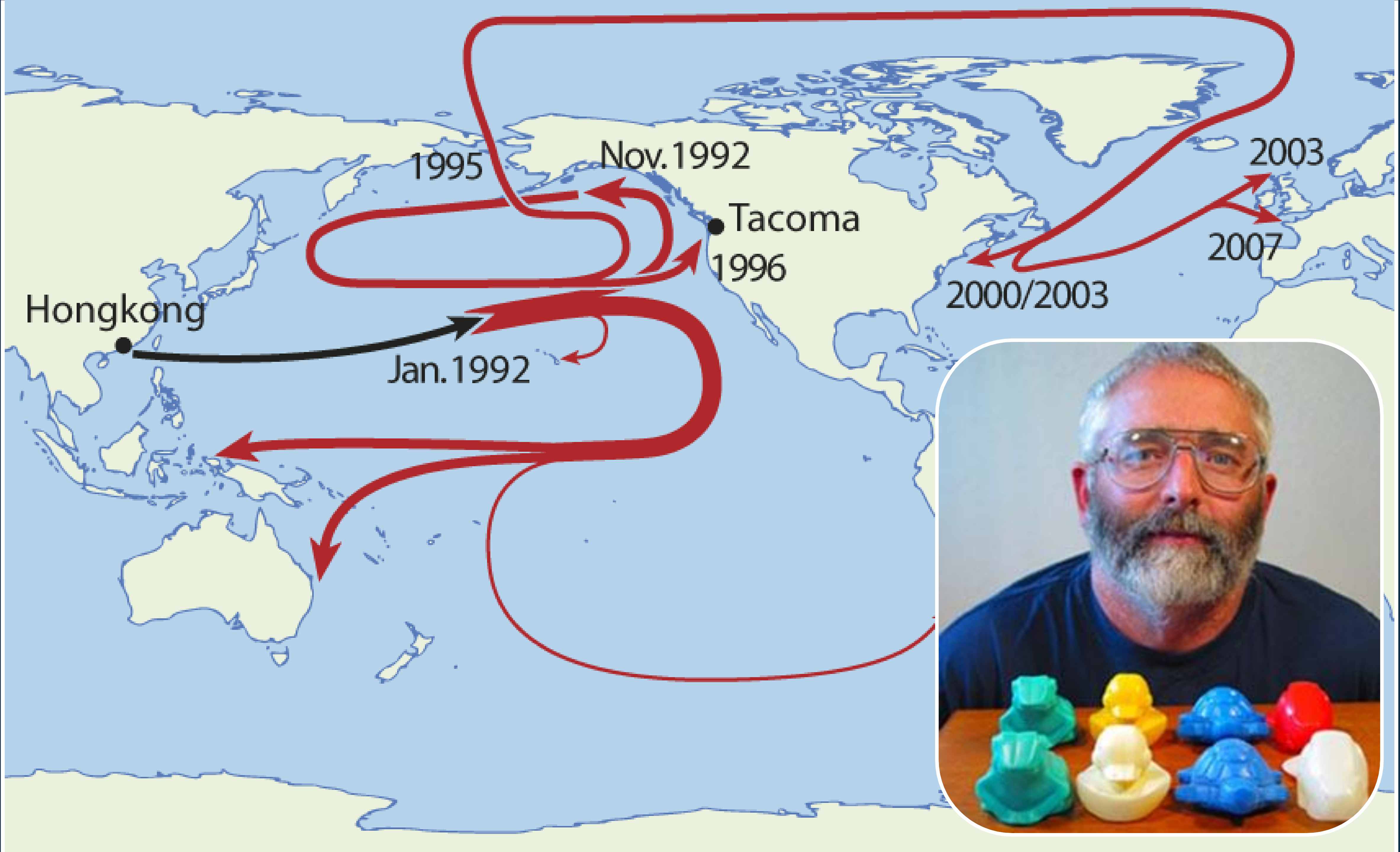
Discussion

Take 5 minutes to discuss in small groups

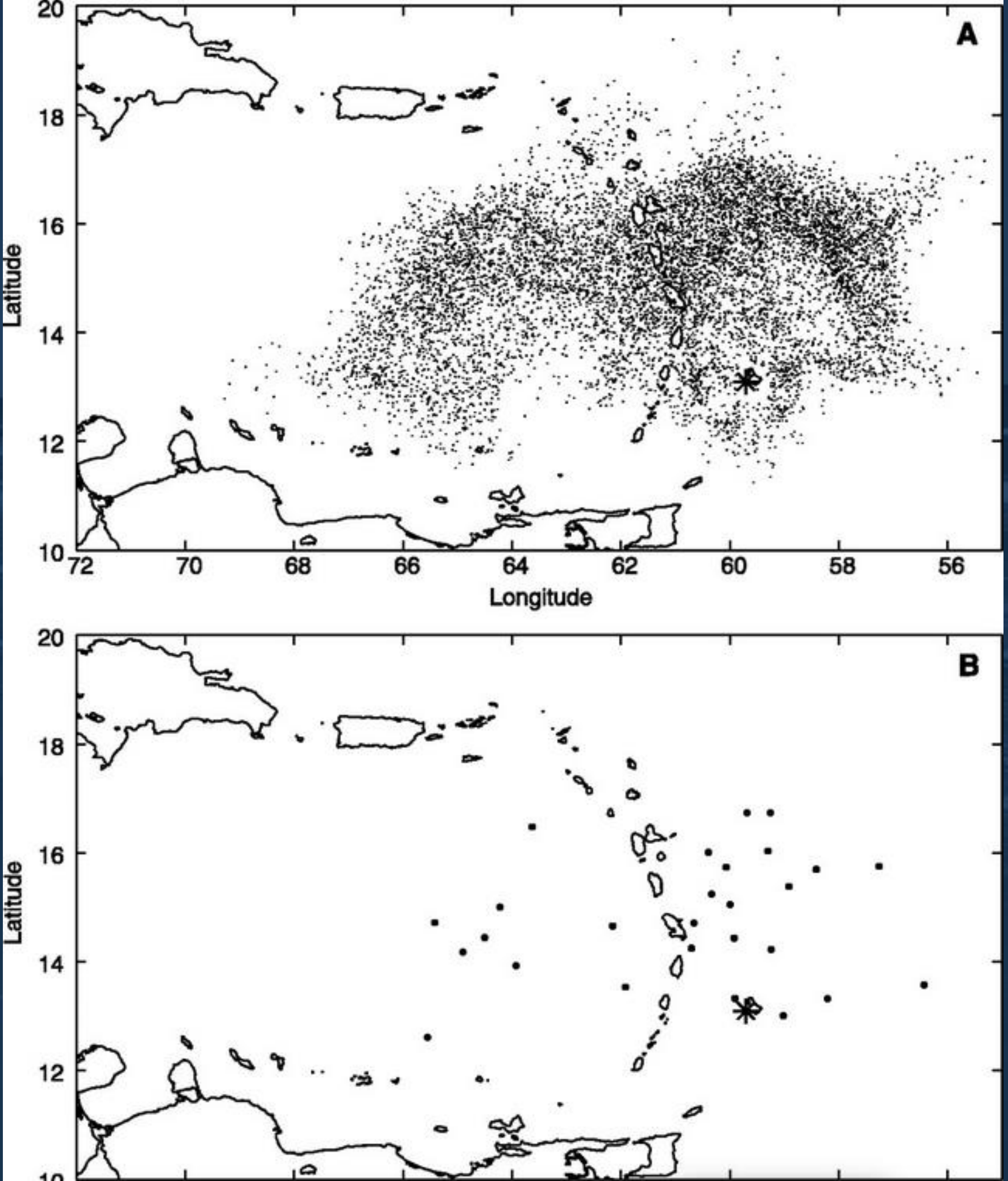
What are three methods that we can use to study actual dispersal in marine larvae (not dispersal potential)?

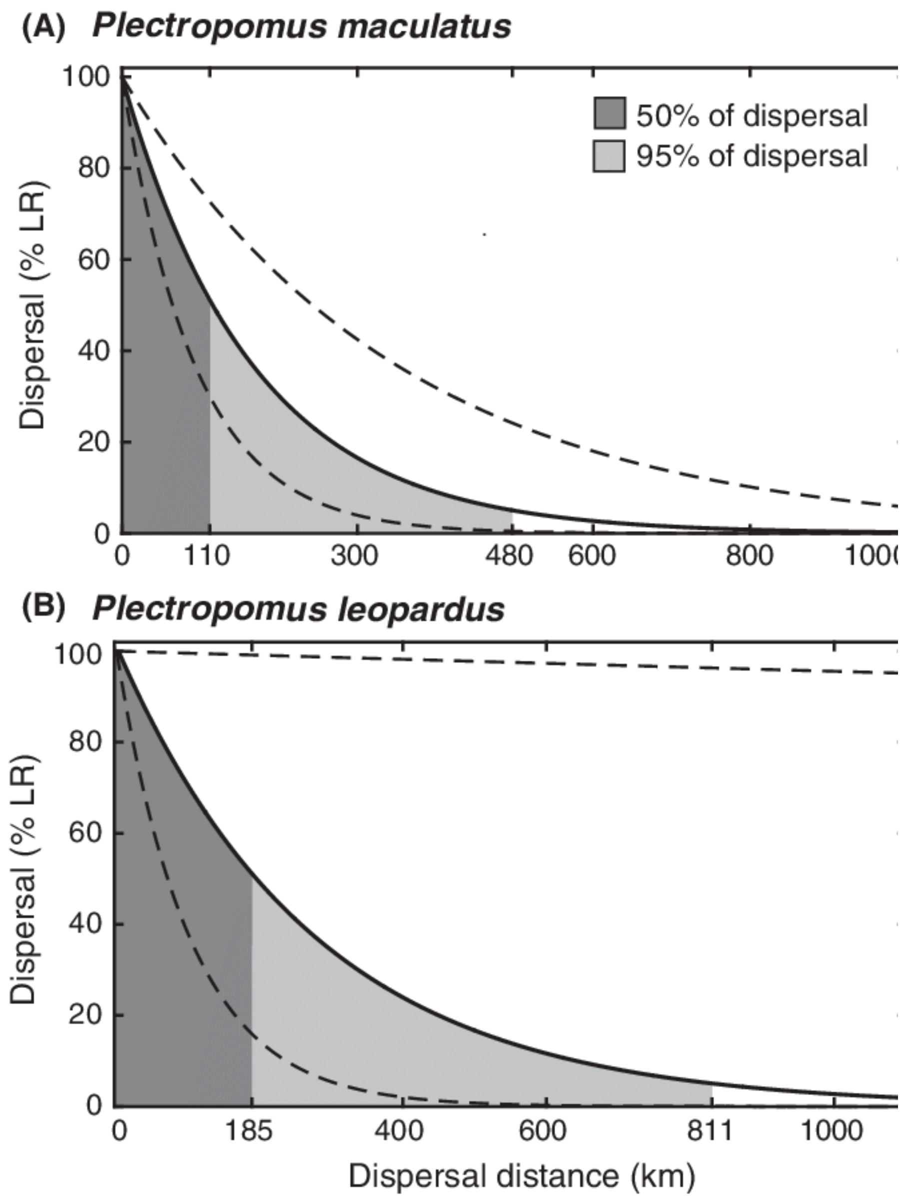



EVERGREEN
EVER GLORY

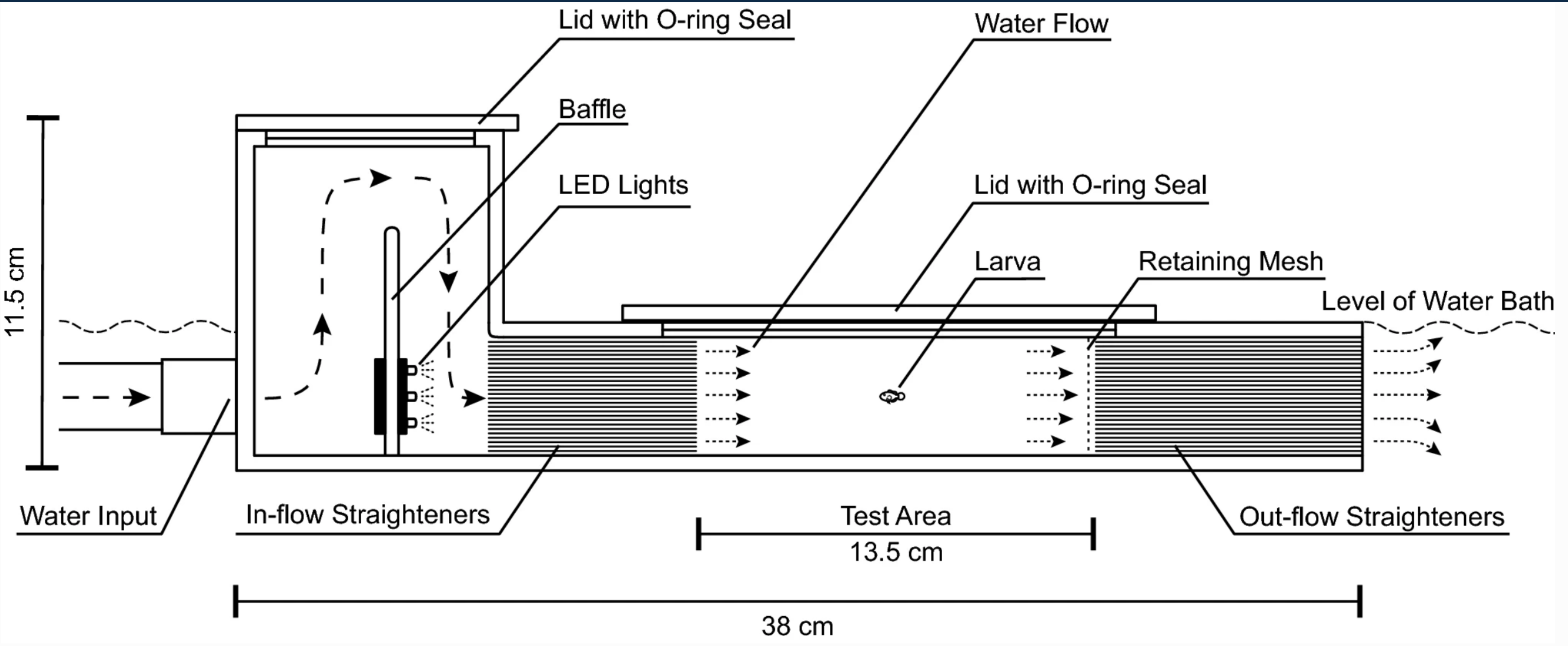

1995
Nov.1992
Tacoma
1996
2003
2007
2000/2003
Hongkong
Jan.1992

Cowen et al. 2000, Science

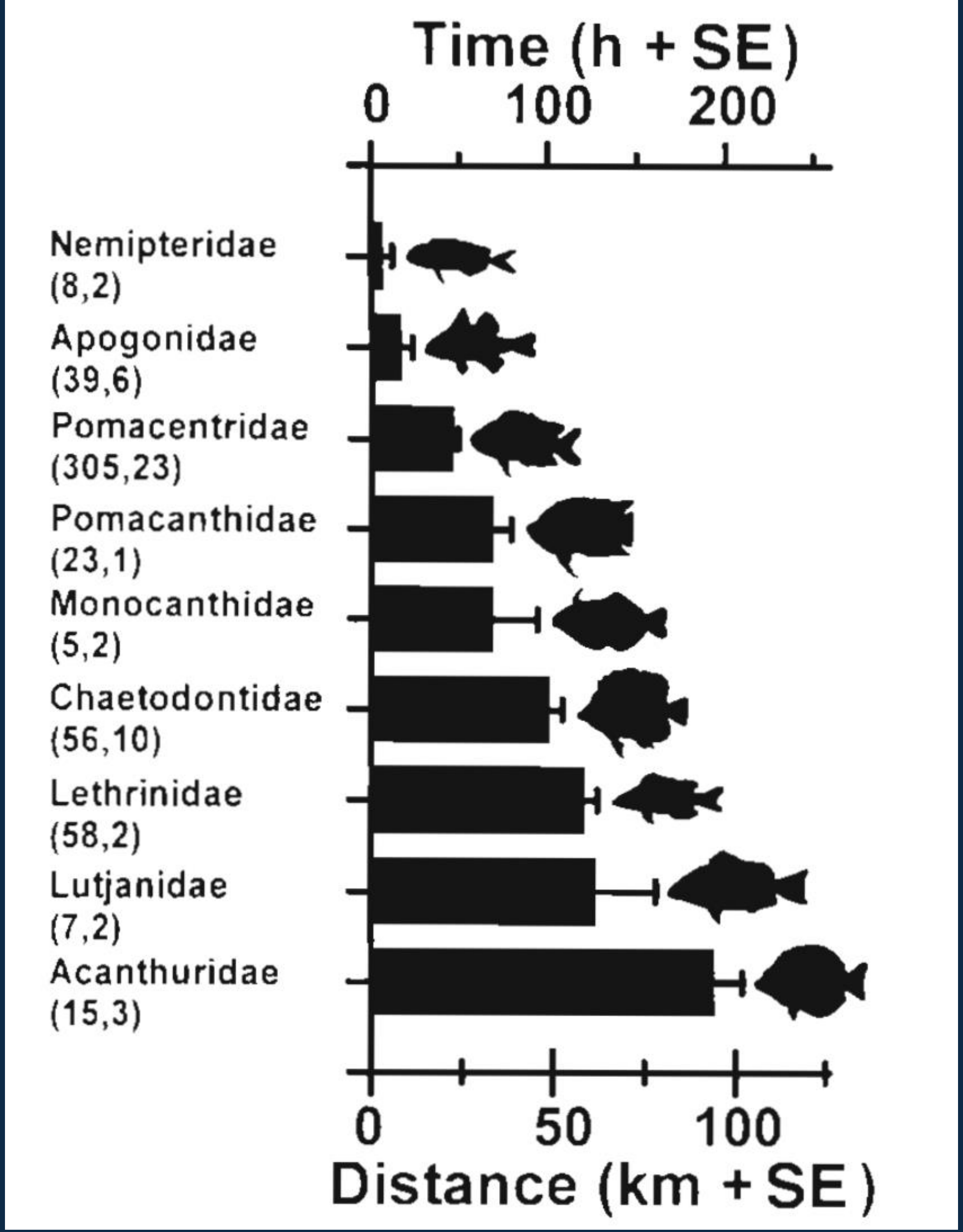




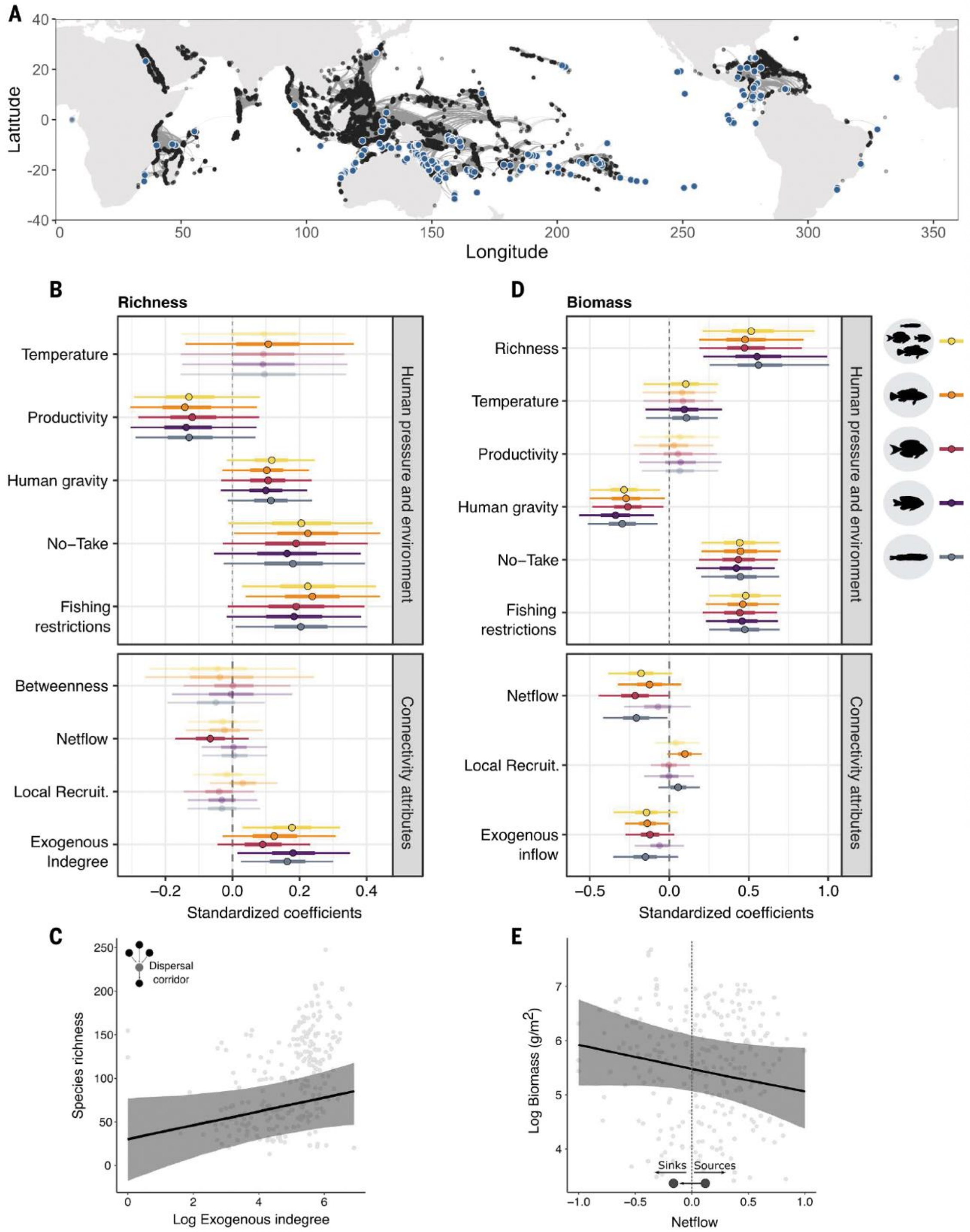
Williamson et al. 2006



Majoris et al. 2019



Stobutzki & Bellwood 1997



RESEARCH

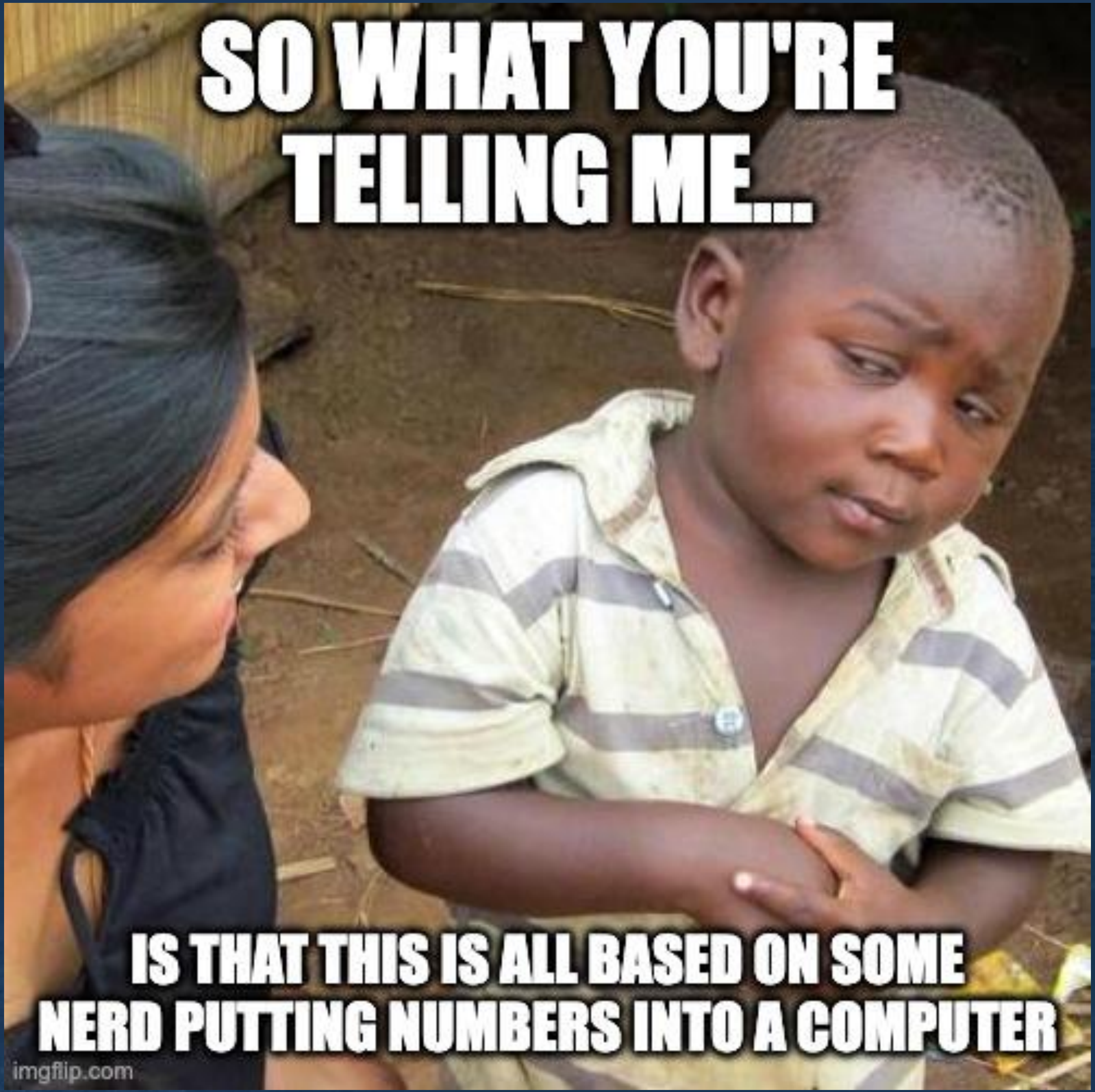
CORAL REEFS

Protecting connectivity promotes successful biodiversity and fisheries conservation

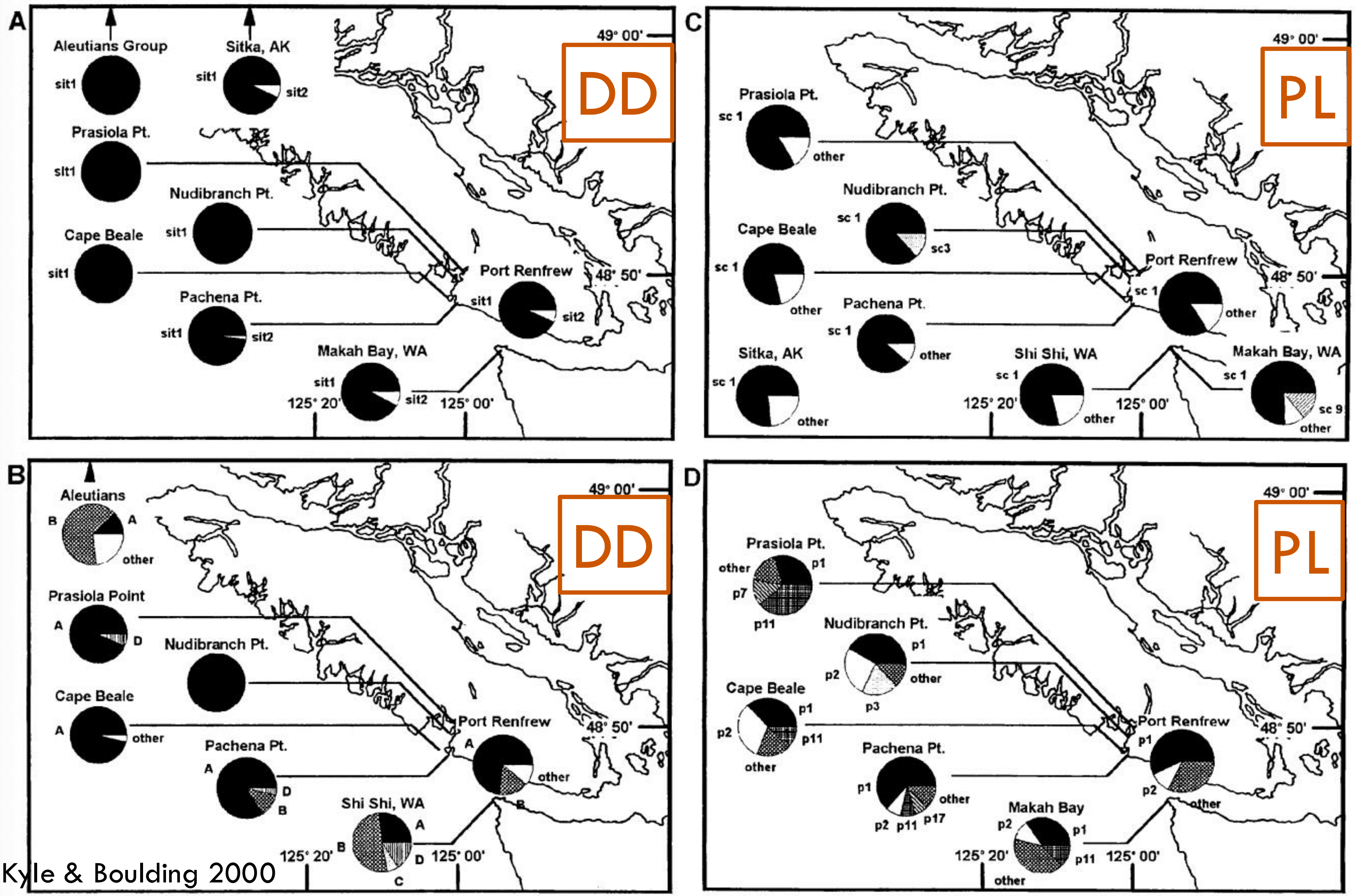
Luisa Fontoura[1], Stephanie D'Agata[1,2,3], Majambo Gamoyo[4], Diego R. Barneche[5,6], Osmar J. Luiz[7], Elizabeth M. P. Madin[8], Linda Eggertsen[9], Joseph M. Maina[1,10]*

The global decline of coral reefs has led to calls for strategies that reconcile biodiversity conservation and fisheries benefits. Still, considerable gaps in our understanding of the spatial ecology of ecosystem services remain. We combined spatial information on larval dispersal networks and estimates of human pressure to test the importance of connectivity for ecosystem service provision. We found that reefs receiving larvae from highly connected dispersal corridors were associated with high fish species richness. Generally, larval "sinks" contained twice as much fish biomass as "sources" and exhibited greater resilience to human pressure when protected. Despite their potential to support biodiversity persistence and sustainable fisheries, up to 70% of important dispersal corridors, sinks, and source reefs remain unprotected, emphasizing the need for increased protection of networks of well-connected reefs.

Fontoura et al. 2022, Science

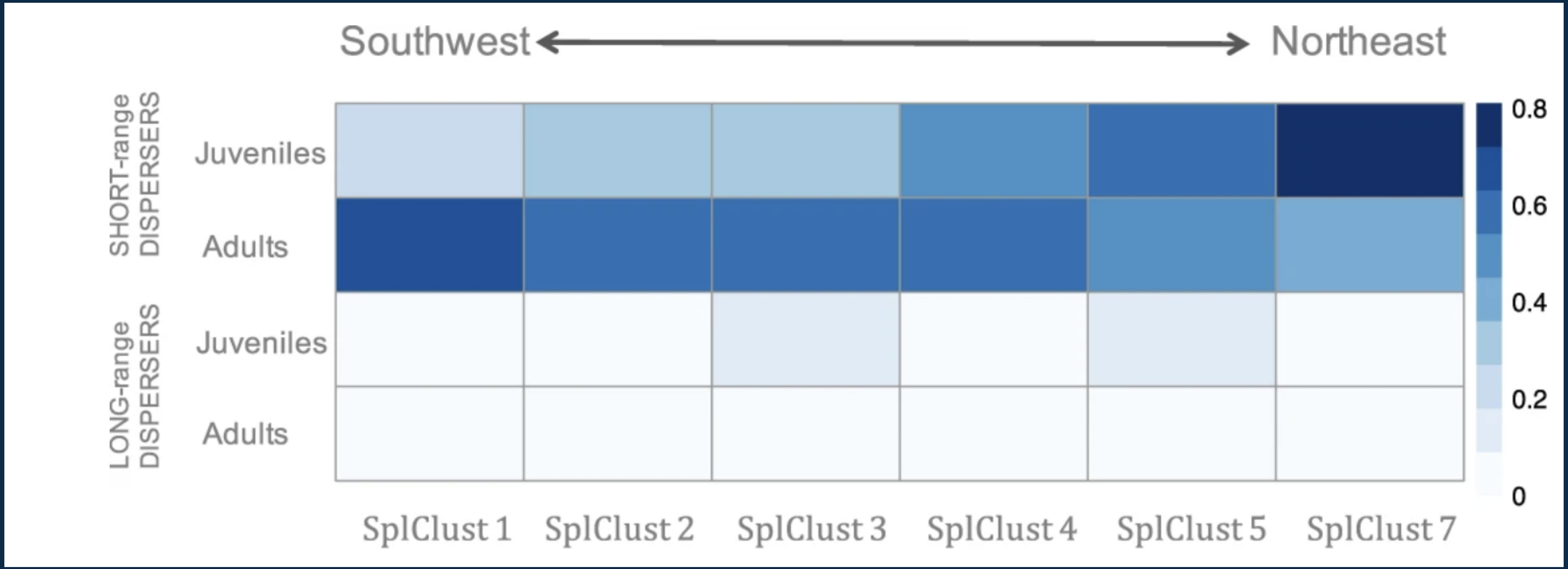

SO WHAT YOU'RE TELLING ME...
IS THAT THIS IS ALL BASED ON SOME NERD PUTTING NUMBERS INTO A COMPUTER
imgflip.com


Kyle & Boulding 2000



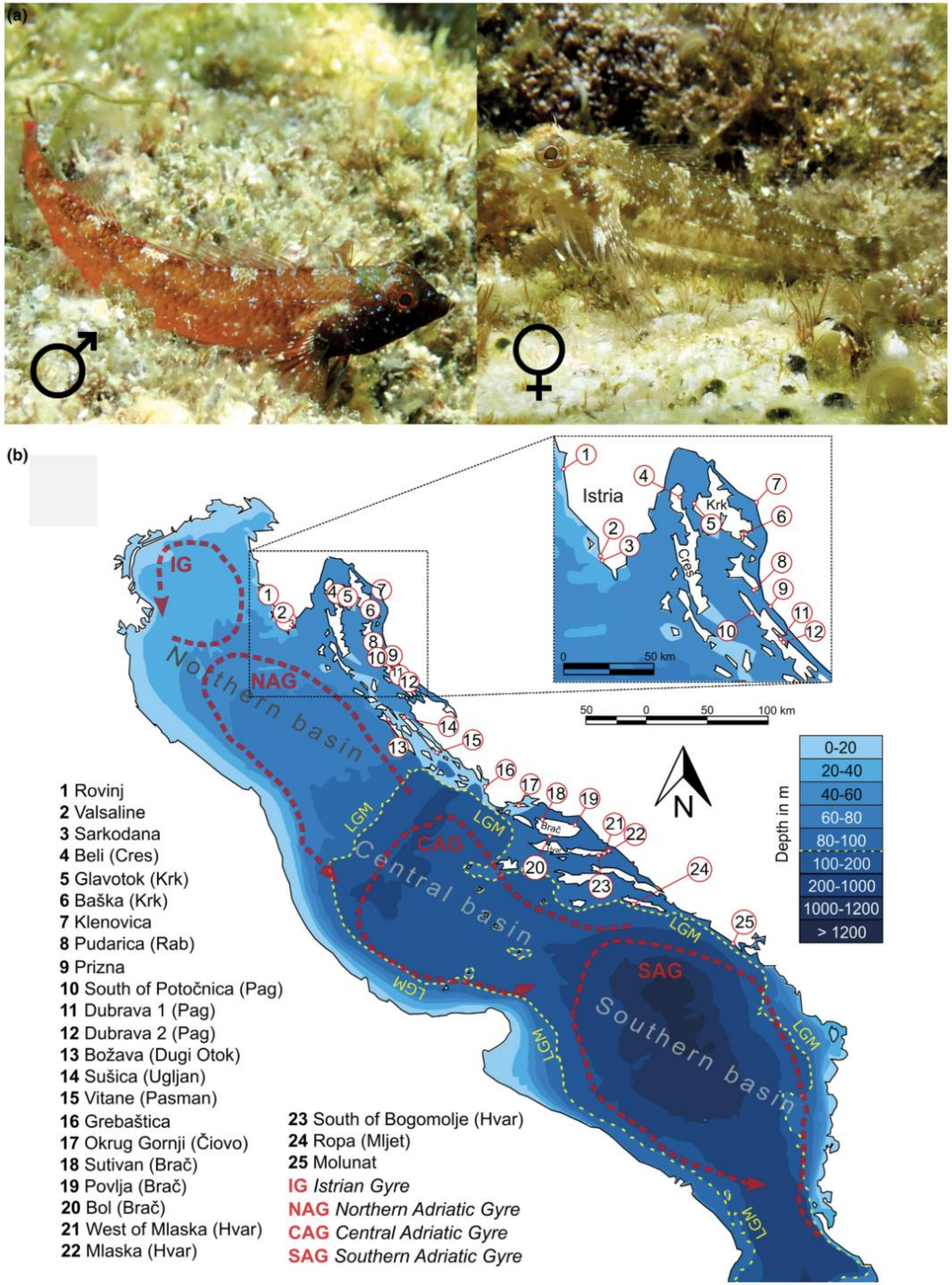
Kyle & Boulding 2000

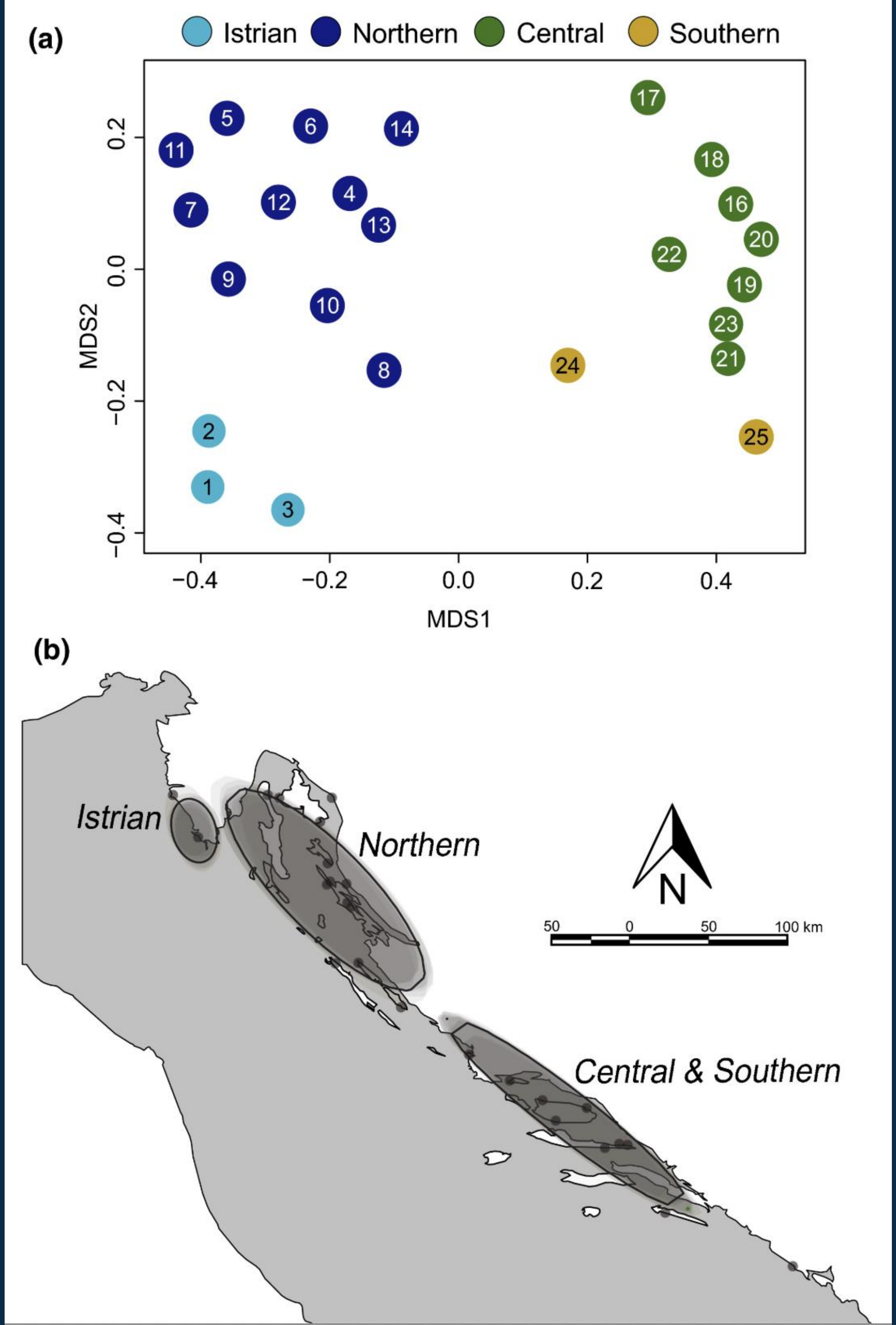
Schunter et al. 2020

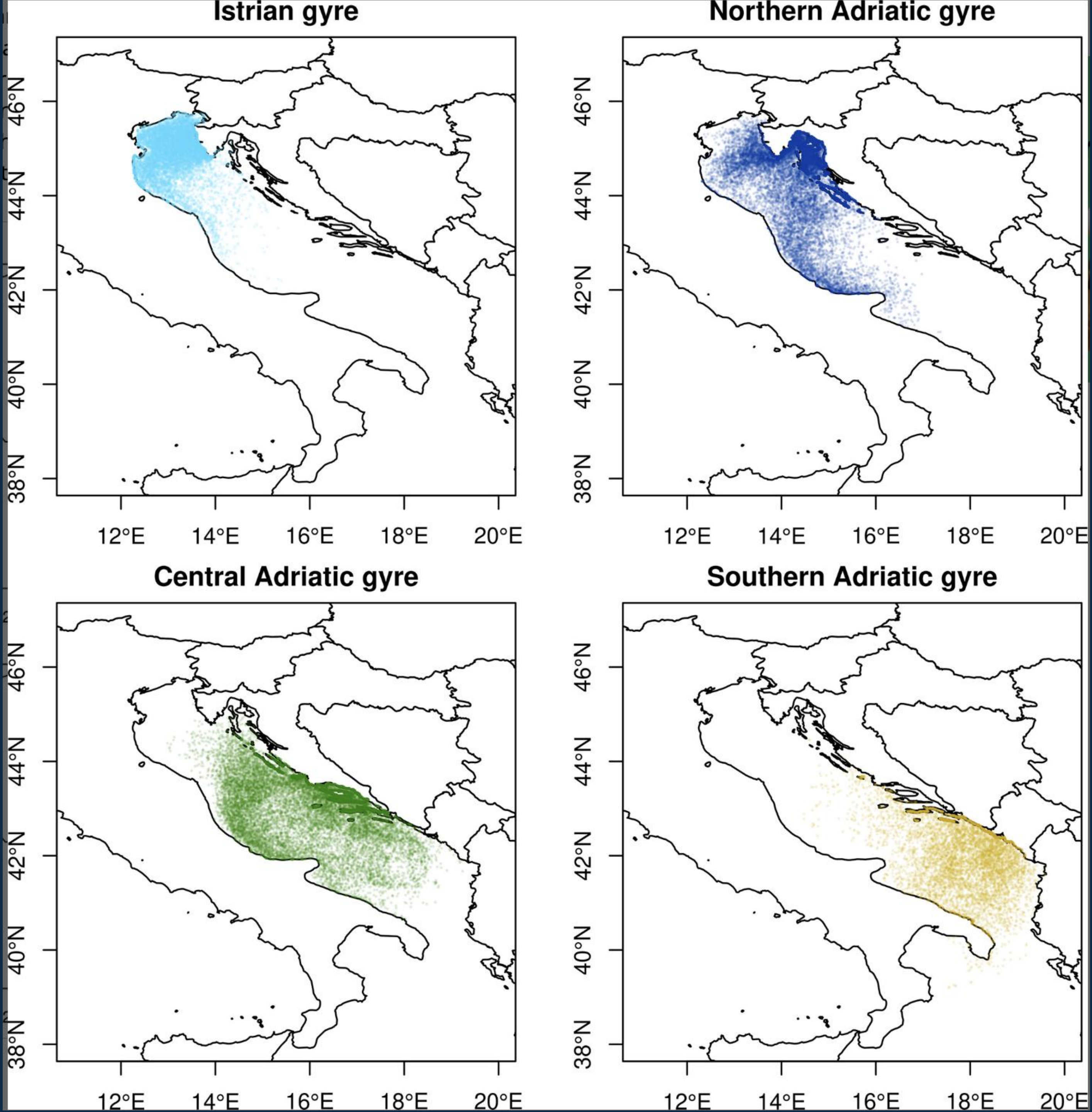


Schunter et al. 2020

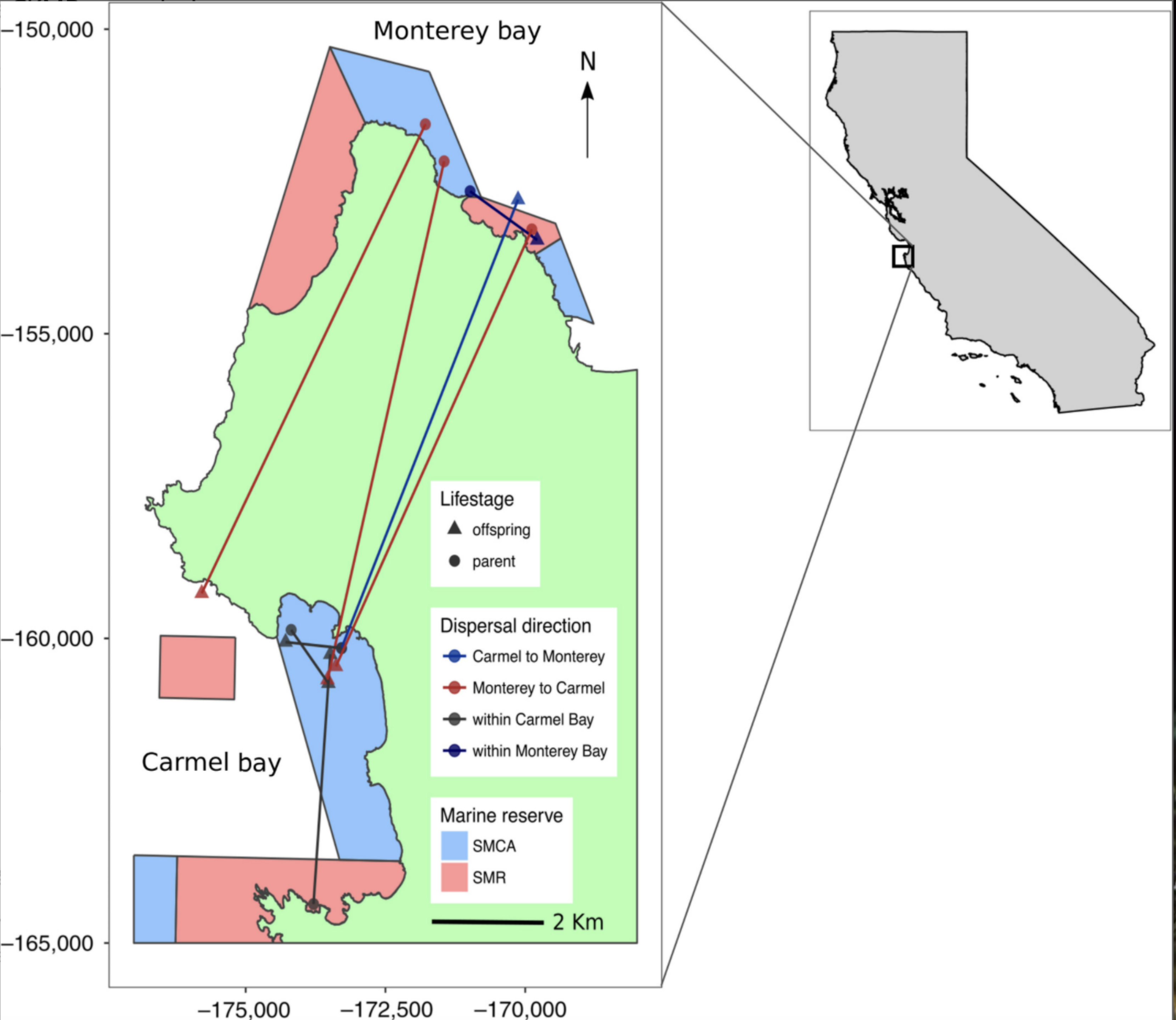
Sefc et al. 2020, J Biogeography



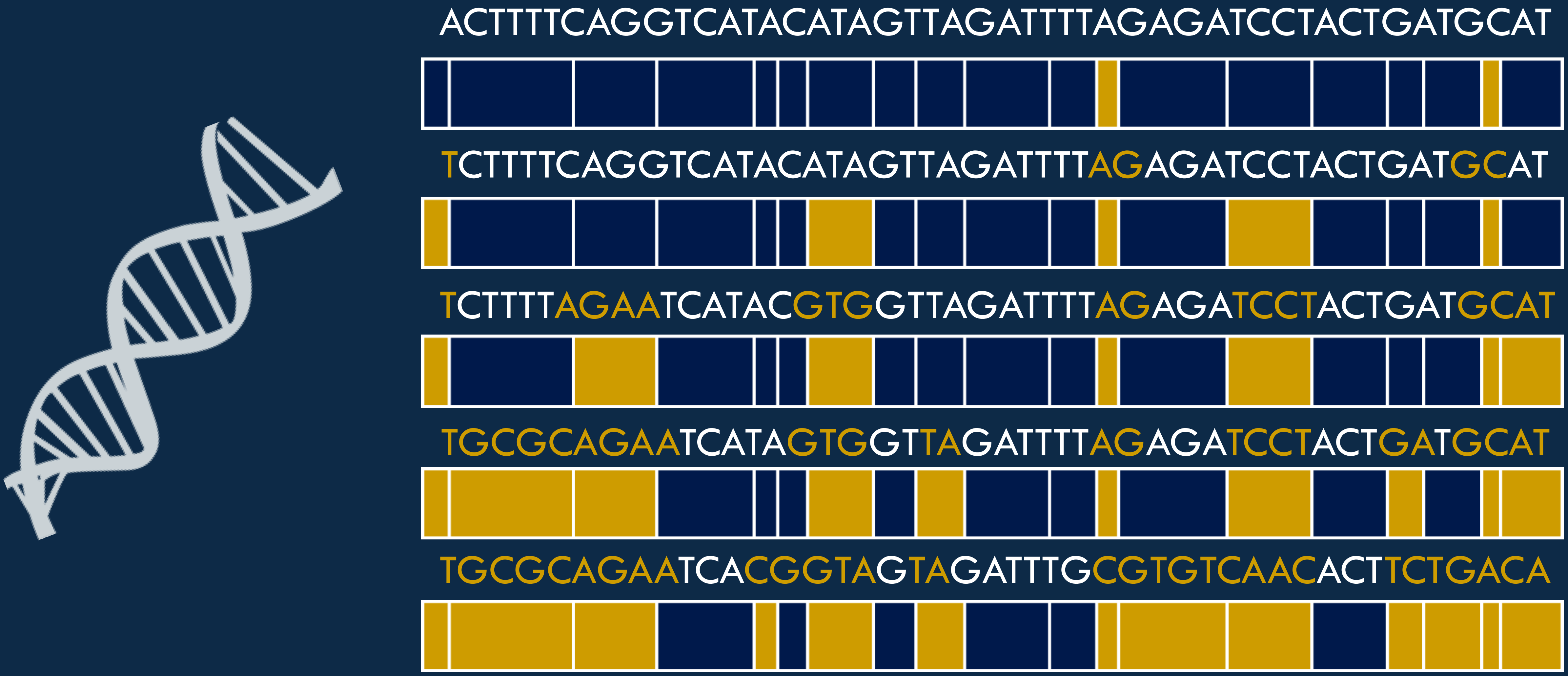


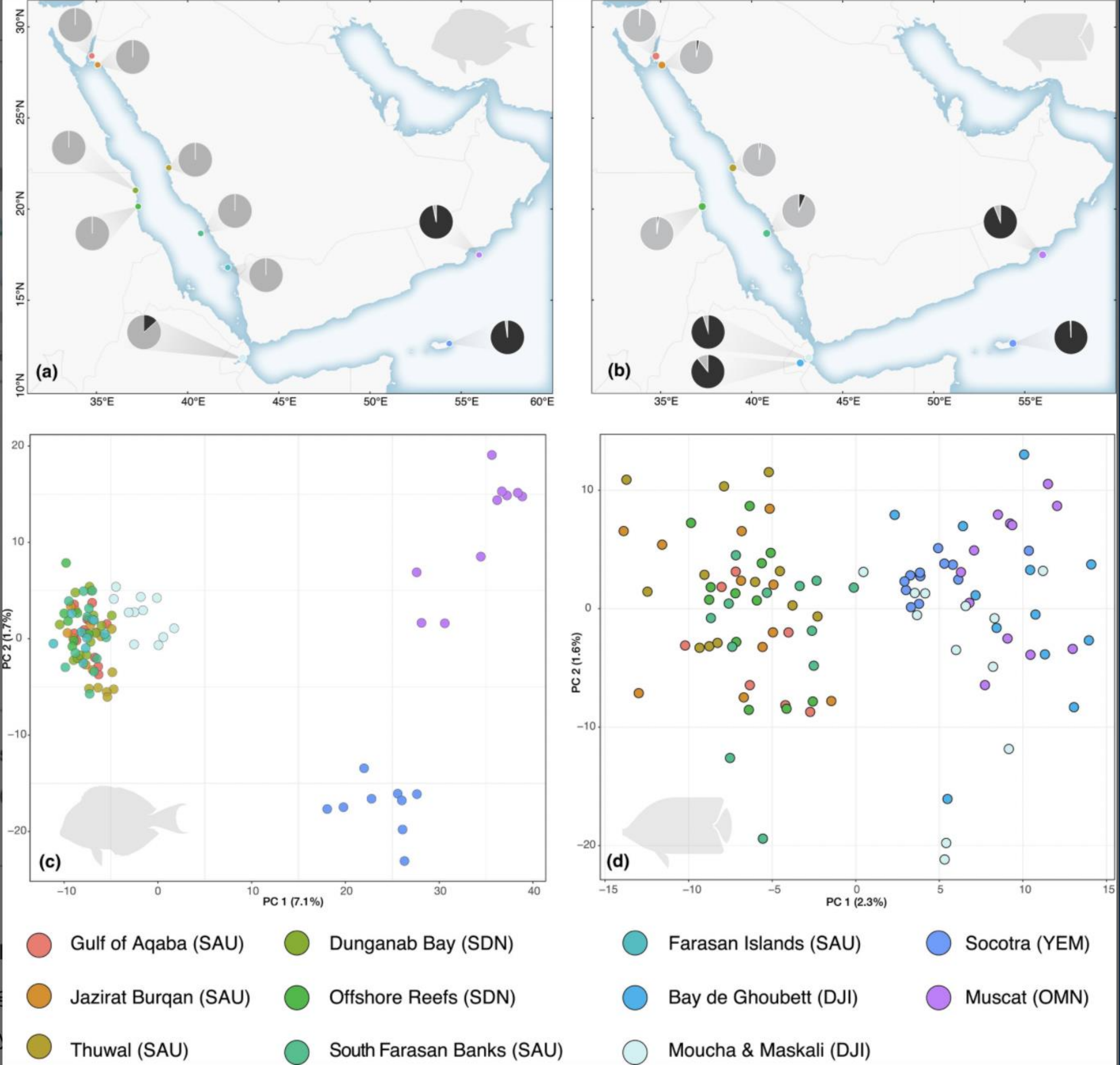


Sefc et al. 2020, J Biogeography

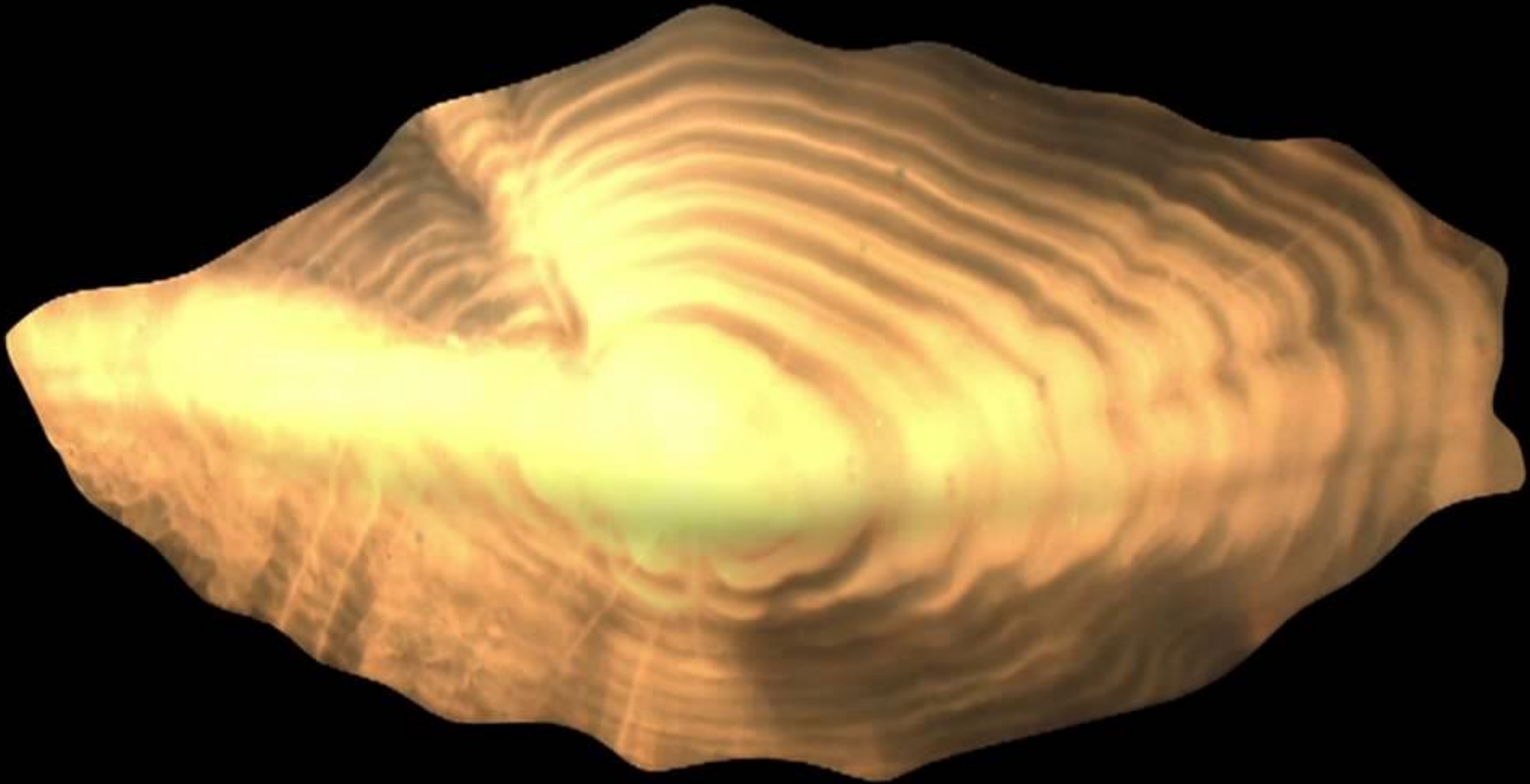


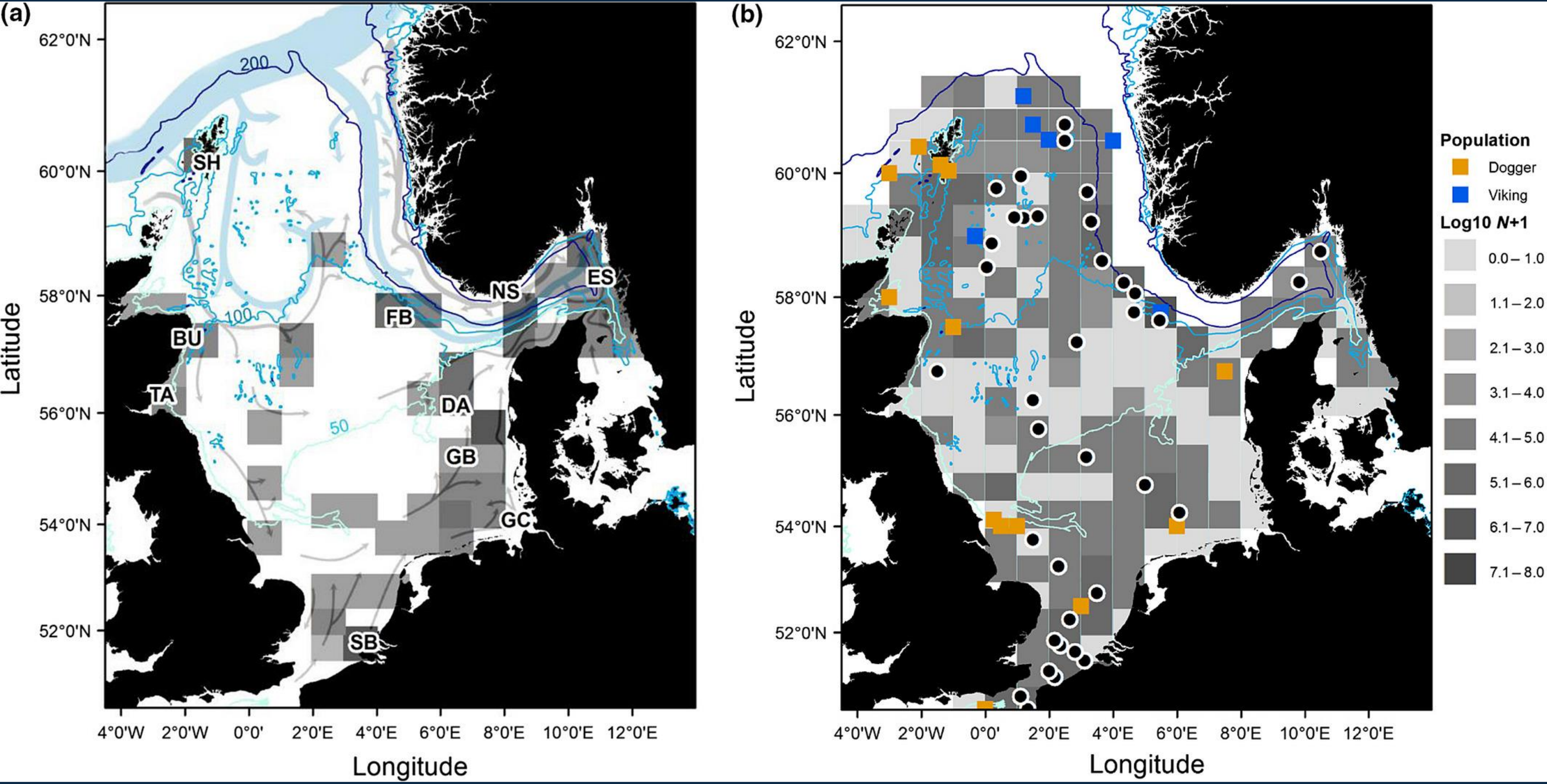
Baetscher et al. 2019


ACTTTTCAGGTCATACATAGTTAGATTTTAGAGATCCTACTGATGCAT
TCTTTTCAGGTCATACATAGTTAGATTTTAGAGATCCTACTGATGCAT
TCTTTTAGAATCATACGTGGTTAGATTTTAGAGATCCTACTGATGCAT
TGCGCAGAATCATAGTGGTTAGATTTTAGAGATCCTACTGATGCAT
TGCGCAGAATCACGGTAGTAGATTTGCGTGTCAACACTTCTGACA



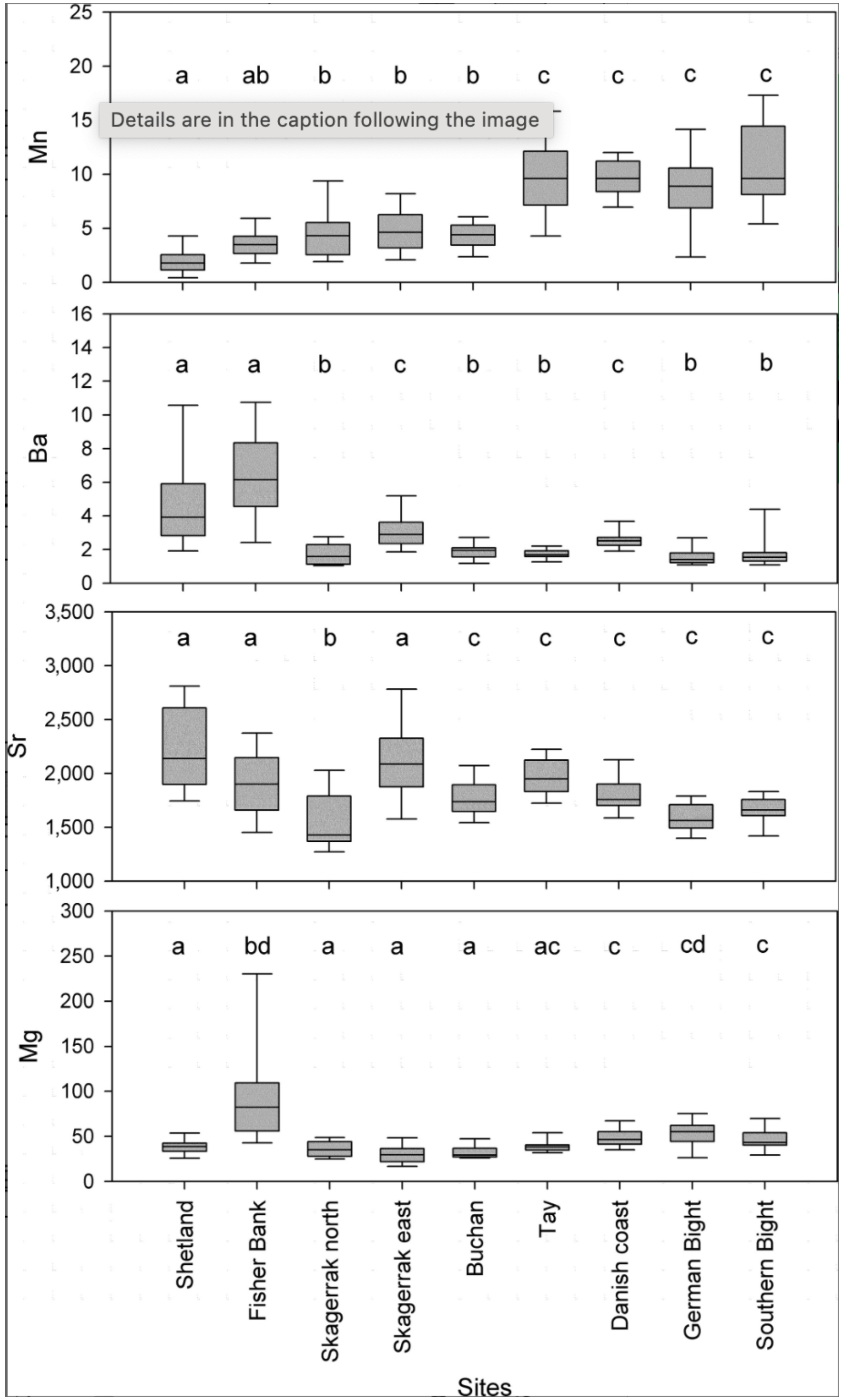
DiBattista et al. 2020

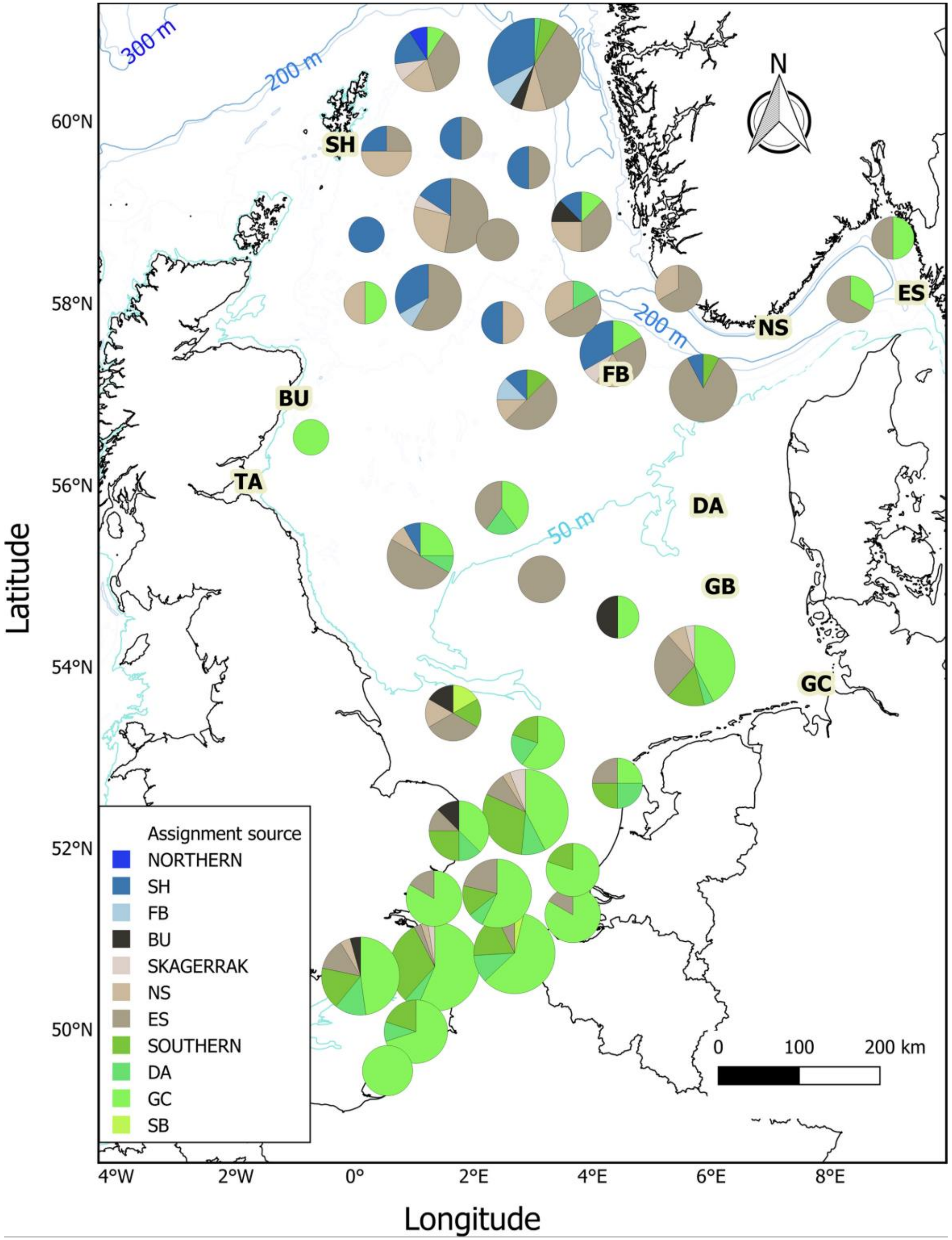




(a)
(b)
62°0'N
60°0'N
58°0'N
56°0'N
54°0'N
52°0'N
Latitude
200
100
50
SH
BU
TA
FB
NS
ES
DA
GB
GC
SB
4°0'W
2°0'W
0°0'
2°0'E
4°0'E
6°0'E
8°0'E
10°0'E
12°0'E
Longitude
Population
Dogger
Viking
Log10 N+1
0.0 – 1.0
1.1 – 2.0
2.1 – 3.0
3.1 – 4.0
4.1 – 5.0
5.1 – 6.0
6.1 – 7.0
7.1 – 8.0

Wright et al. 2018, Ecol Evo






25 G
T17000-25.0
RPI
RESEARCH
PRODUCTS
INTERNATIONAL
Tetracycline HCl
$C_{22}H_{24}N_2O_8 \cdot HCl$
MW 480.90
CAS No. 64-75-5
Soluble in water.
Store at RT
Lot 22363-43393
For Research and Further Manufacturing Use Only, Not for Food or
www.rpicorp.com 800-323-9814 ISO 9001:2015 & ISO 22716:2007

1,926 clutches – 10 million embryos


5,000 juveniles
15 tetracycline rings
reeflifesurvey.com

Jones et al. 1999, Nature

Kimbe Bay, Papua New Guinea

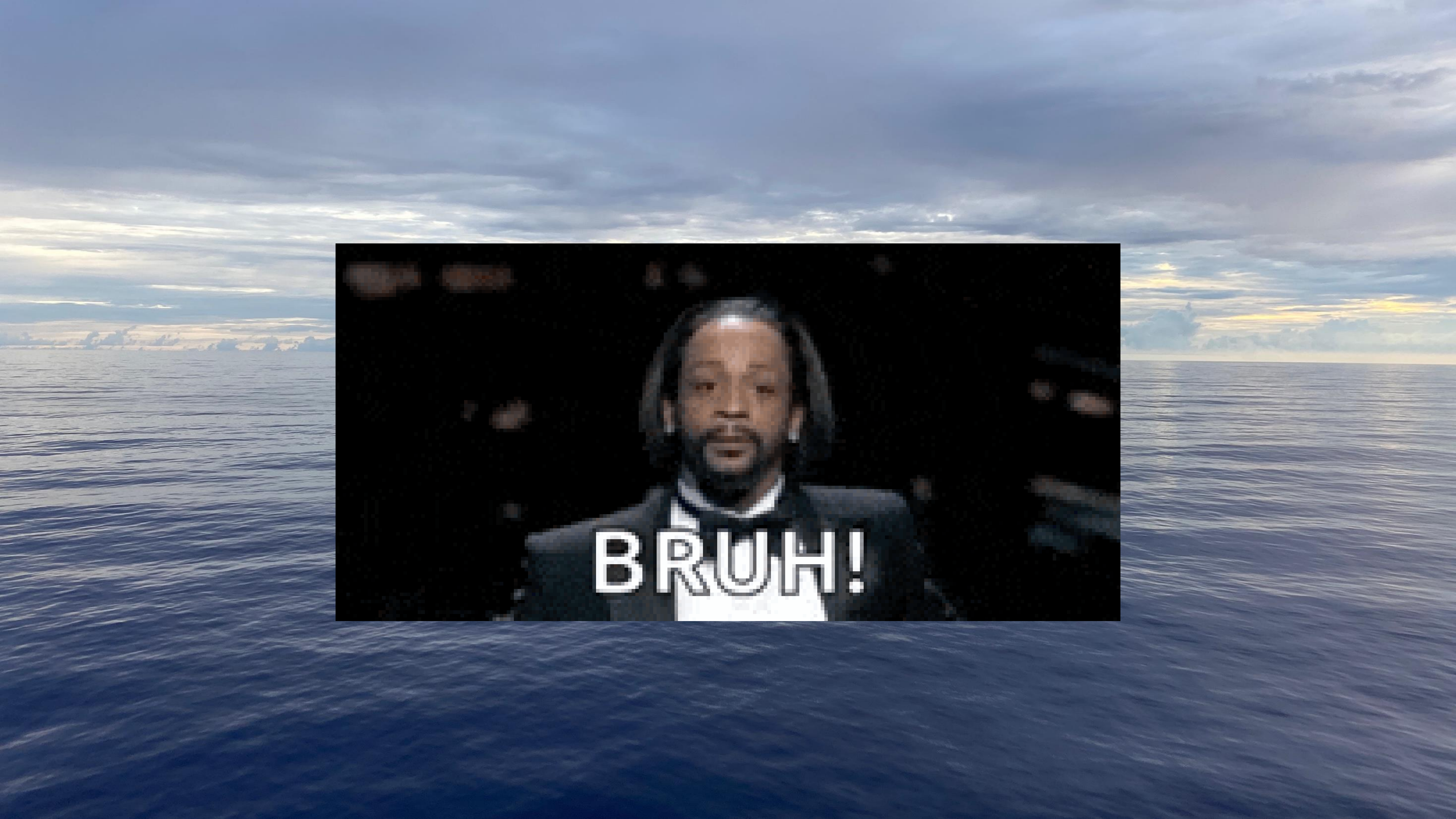
Plectropomus areolatus

Amphiprion percula

Chaetodon vagabundus

Almany et al. 2007, Science

Seriatopora hystrix


BRUH!

NEXT CLASS: SPECIATION